# PHYSICS

1) Three particles A, B and C are situated at the vertices of an equilateral triangle of sides 'd' at t = 0. Each of the particles moves with constant speed 'v'. 'A' always has its velocity along AB, 'B' along BC and 'C' along CA. At what time will the particles meet each other :-

(1) $\frac{2d}{3v}$

(2) $\frac{d}{3v}$

(3) $\frac{3d}{2v}$

(4) $\frac{4d}{3v}$

2) A river is flowing from west to east at a speed of 15 metres per minute. A man on the south bank of the river capable of swimming at 30 metres per minute in still water, wants to swim across the river in shortest time. He should swim in a direction

(1) Due north

(2) 30° east of north

(3) 30° west of north

(4) 60° east of north

3) A lift is moving with a uniform downward acceleration of $2ms^{-2}$. A ball is dropped from a height 2m from the floor of lift. Find the time taken after which ball will strike the floor (in s) :-

(1) $\sqrt{2}$

(2) 2

(3) $1/\sqrt{2}$

(4) 1/2

4) Two particles A and B are projected simultaneously in the directions shown in figure with velocities $v_A$ = 25 m/s and $v_B$ = $10\sqrt{3}$ m/s. If they collide in air after 2s, the angle θ is :




(1) 30°

(2) 45°

(3) 53°

(4) 37°

5) A man is walking on a road with a velocity of 3km/h suddenly starts raining and velocity of rain is 10km/h in vertically downward direction, then relative velocity of the rain with respect to man is:

(1) $\sqrt{13}$ km/hr
(2) $\sqrt{7}$ km/hr
(3) $\sqrt{109}$ km/hr
(4) 13 km/h

6) A projectile is projected with kinetic energy K. If it has the maximum possible horizontal range, then its kinetic energy at the highest point will be :-

(1) 0.25 K
(2) 0.5 K
(3) 0.75 K
(4) 1.0 K

7) A car with a vertical windshield moves in a rain storm at a speed of 40 km/h. The rain drops fall vertically with constant speed of 20 m/s. The angle at which rain drops strike the windshield is :-

(1) $\tan^{-1}\left(\frac{5}{9}\right)$
(2) $\tan^{-1}\left(\frac{9}{5}\right)$
(3) $\tan^{-1}\left(\frac{3}{2}\right)$
(4) $\tan^{-1}\left(\frac{2}{3}\right)$

8)

When the angle of elevation of a gun are 60° and 30° respectively, the maximum heights it shoots are $h_1$ and $h_2$ respectively, $h_1/h_2$ equal to (same initial speed in both cases) -

(1) 3/1
(2) 1/3
(3) 1/2
(4) 2/1

9) A bomber is flying horizontally with a constant speed of 150 m/s at a height of 78.4 m. The pilot has to drop a bomb at the enemy target. At what horizontal distance from the target should he release the bomb?

(1) 0 m
(2) 300 m
(3) 600 m
(4) 1000m

10) Particle is projected upward with velocity 100 m/s at an angle of 30° with horizontal. Find the time when its velocity is perpendicular to the acceleration :-

(1) 10 s

(2) 5 s

(3) 20 s

(4) 15 s

11) A person walks up a stalled escalator in 90 sec. He is carried in 60s, when standing on the same escalator which is now moving. The time he would take to walk up the moving escalator will be :–

(1) 27 s

(2) 72 s

(3) 18 s

(4) 36 s

12) Two cars A and B start moving from the same point with same speed 36 km/h. Car A moves towards North and car B moves towards East. What is the relative velocity of B w.r.t. A :

(1) $36\sqrt{2}$m/s towards South-East

(2) $36\sqrt{2}$m/s towards North-West

(3) $36\sqrt{2}$m/s towards North-East

(4) None of these

13) A particle is projected under gravity at an angle of projection 45°. If for ground to ground projectile motion its range is 36 m. Then maximum height attained by particle : –

(1) 6 m

(2) 9 m

(3) 5 m

(4) 8 m

14) A ball having initial velocity as $\vec{v} = 20\hat{i} + 40\hat{j}$ , calculate time of flight?

(1) 4 sec

(2) 8 sec

(3) 16 sec

(4) 5 sec

15) A ball is projected with a velocity, 10 $ms^{-1}$, at an angle of 60° with the vertical direction. Its speed at the highest point of its trajectory will be :

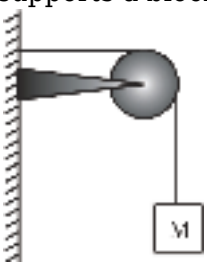
(1) $5\sqrt{3}\ ms^{-1}$

(2) 5 $ms^{-1}$

(3) 10 $ms^{-1}$

(4) Zero

16) A string of negligible mass going over a clamped pulley of mass m supports a block of mass M as shown in the figure. The force on the pulley by the clamp is given by :-



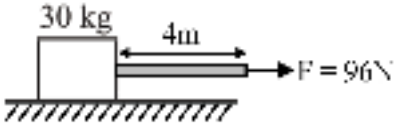

(1) $\sqrt{2}$ Mg

(2) $\sqrt{2}$ mg

(3) $\left(\sqrt{(M+m)^2+m^2}\right)g$

(4) $\left(\sqrt{(M+m)^2+M^2}\right)g$

17) A 30 kg body is pulled by a rope 4 m long on a frictionless horizontal surface by a force 96 N. If mass of the rope is 2 kg, force acting on the body is :




(1) 96 N

(2) 94 N

(3) 92 N

(4) 90 N

18) A 5000 kg rocket is set for vertical firing. The exhaust speed is 800 $ms^{-1}$. To give an initial upward acceleration of 20 $ms^{-2}$, the amount of gas ejected per second to supply the needed thrust will be (g = 10 $ms^{-2}$) :-

(1) 127.5 kg $s^{-1}$

(2) 187.5 kg $s^{-1}$

(3) 200 kg $s^{-1}$

(4) 137.5 kg $s^{-1}$

19) Two bodies of mass 3 kg and 4 kg are suspended at the ends of massless string passing over a frictionless pulley. The acceleration of the bodies is (g = 9.8 $m/s^2$) :-

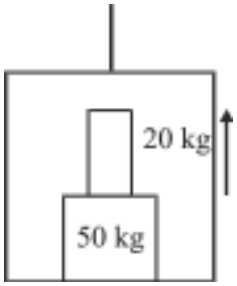
(1) 4.9 $m/s^2$

(2) 2.45 $m/s^2$

(3) 1.4 $m/s^2$

(4) 9.5 $m/s^2$

20) On the floor of an elevator, a block of mass 50 kg is placed, on which another block of mass 20 kg is also placed. The elevator is moving up with a constant acceleration 1.5 $m/s^2$. Force exerted by

20 kg block on the 50 kg block is nearly :- 


(1) 250 N

(2) 230 N

(3) 170 N

(4) 150 N

21) A cricket ball of mass 150 g is moving with a velocity of 12m/sec and is hit by a bat so that the ball is turned back with a velocity of 20 m/sec, the force on the ball acts for 0.01 sec, then the average force exerted by the bat on the ball will be

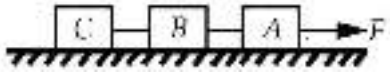
(1) 48 N

(2) 40 N

(3) 480 N

(4) 400 N

22) Three identical blocks of masses m = 2 kg are drawn by a force F = 10.2 N on a frictionless surface, then what is the tension (in N) in the string between the blocks B and C ?

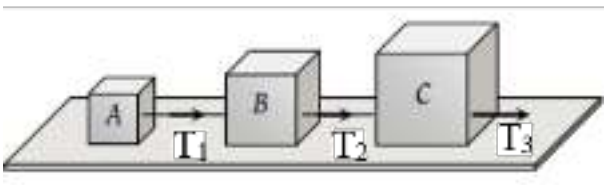



(1) 9.2

(2) 3.4

(3) 4

(4) 9.8

23) Three blocks A, B and C weighing 1, 8 and 27 kg respectively are connected as shown in the figure with an inextensible string and are moving on a smooth surface , $T_3$ is equal to 36 N. Then $T_2$ is : - 

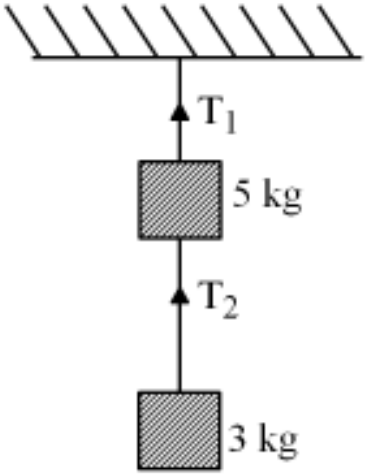

(1) 18 N

(2) 9 N

(3) 3.375 N

(4) 1.25 N

24) Two masses of 5 kg and 3 kg are suspended with help of massless inextensible strings as shown in figure. Calculate $T_1$ and $T_2$ when whole system is going upwards with acceleration = 2 m $s^{-2}$ (use g




= 9.8 m $s^{-2}$).

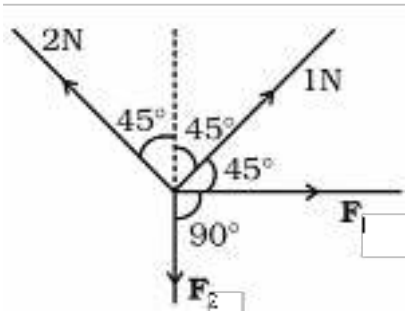
(1) $T_1$ = 95.4 N, $T_2$ = 36.4 N

(2) $T_1$ = 93.6 N, $T_2$ = 33.4 N

(3) $T_1$ = 94.4 N, $T_2$ = 35.4 N

(4) $T_1$ = 90.5 N, $T_2$ = 33.3 N

25) There are four forces acting at a point P produced by strings as shown in figure which is at rest.




Find the forces $\mathbf{F}_1$ and $\mathbf{F}_2$.

(1) $\frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}$

(2) $\sqrt{2}, \frac{2}{\sqrt{3}}$

(3) $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{3}}$

(4) $\frac{1}{\sqrt{2}}, \frac{3}{\sqrt{2}}$

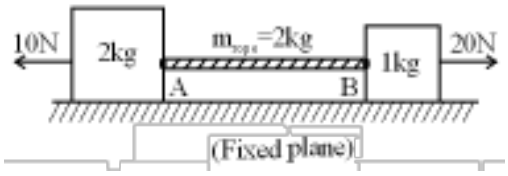
26) A person of mass 50 kg stands on a weighing scale on a lift. If the lift is descending with a downward acceleration of 9 m $s^{-2}$, what would be the reading of the weighing scale? (g = 10 m $s^{-2}$)

(1) 6 kg

(2) 4 kg

(3) 3 kg

(4) 5 kg

27) 

Find tension at point A and point B of rope as shown. [Given floor is smooth]

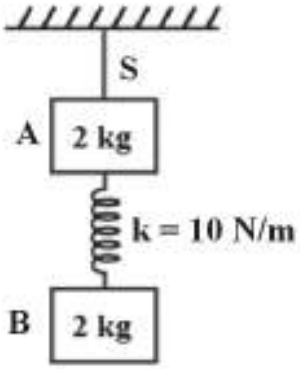
(1) $T_A$ = 18 N, $T_B$ = 14 N

(2) $T_A$ = 14 N, $T_B$ = 18 N

(3) $T_A$ = 4 N, $T_B$ = 18 N

(4) $T_A$ = 4 N, $T_B$ = 8 N

28) The system shown in figure is in equilibrium. The spring is light, the acceleration of both the




blocks (in $m/s^2$) just after the string S is cut is :

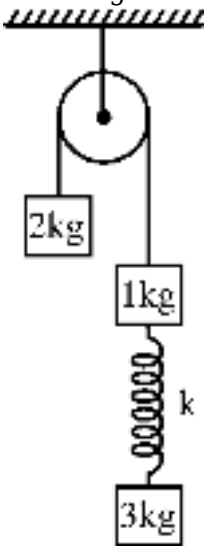
(1) $a_A$ = 10, $a_B$ = 0

(2) $a_A$ = 0, $a_B$ = 10

(3) $a_A$ = 20, $a_B$ = 0

(4) $a_A$ = 0, $a_B$ = 20

29) From the fixed pulley, masses 2kg, 1kg and 3kg are suspended as shown in the figure. Find the



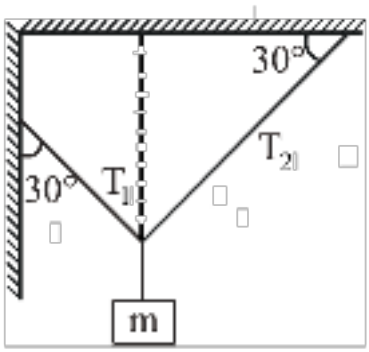

extension in the spring if k = 100 N/m. (Neglect oscillations due to spring) :-

(1) 0.1 m

(2) 0.2 m

(3) 0.3 m

(4) 0

30) 

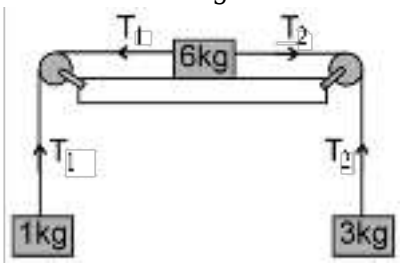

The ratio of $T_1$ and $T_2$ is :-

(1) $\frac{1}{\sqrt{3}}$

(2) $\frac{\sqrt{3}}{1}$

(3) $\frac{2}{\sqrt{3}}$

(4) $\frac{\sqrt{3}}{4}$

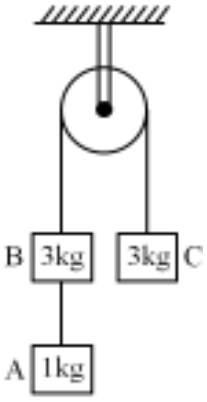
31) Three masses of 1 kg, 6 kg and 3 kg are connected to each other with threads and are placed on table as shown in figure. What is the acceleration with which the system is moving? Take g = 10m $s^{-2}$. 


(1) Zero

(2) 1 $ms^{-2}$

(3) 2 $ms^{-2}$

(4) 3 $ms^{-2}$

32) In the system shown in the figure, the acceleration of the 1 kg mass and the tension in the string between A and B is :- 

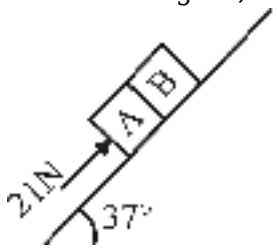
(1) $\frac{g}{4}$ downwards, $\frac{8g}{7}$

(2) $\frac{g}{4}$ upwards, $\frac{g}{7}$

(3) $\frac{g}{7}$ downwards, $\frac{6}{7}g$

(4) $\frac{g}{2}$ upwards, g

33) Blocks A and B of masses 2kg and 1kg respectively are pushed upon a frictionless inclined plane by a 21 N force applied parallel to the inclined plane as shown in figure, the magnitude of the force of normal reaction between A and B (g = 10 m/s$^2$) is :-




(1) 7 N

(2) 5 N

(3) 9 N

(4) 3 N

34) In the given arrangement, find the acceleration of mass 4kg at the instant when acceleration of block of mass 2kg is 2m/s$^2$.




(1) 4m/s$^2$

(2) 2m/s$^2$

(3) 5m/s$^2$

(4) 6m/s$^2$

35) Ten one-rupee coins are put on top of each other on a table. Each coin has a mass m. The reaction of the 4th coin (counted from the bottom) on the 5th coin is :-

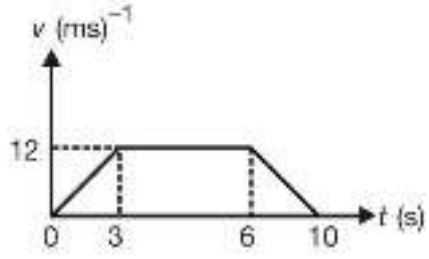
(1) 4 mg

(2) 5 mg

(3) 6 mg

(4) 7 mg

36) A lift is moving in upward direction. The total mass of the lift and the passenger is 1600 kg. The variation of the velocity of the lift is as shown in the figure. The tension in the rope at t = 8 seconds

will be 

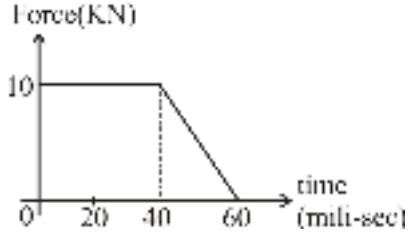

(1) 11200 N

(2) 16000 N

(3) 48000 N

(4) 12000 N

37) A force-time plot for a body is show in the figure. The total change in momentum of the body is -




(1) 400 N-s

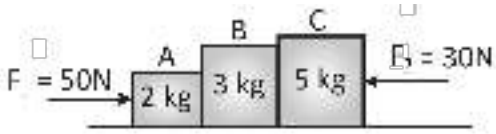
(2) 300 N-s

(3) 500 N-s

(4) 200 N-s

38)

Find contact force between the block - B and block - C as shown in figure : (All surface smooth)



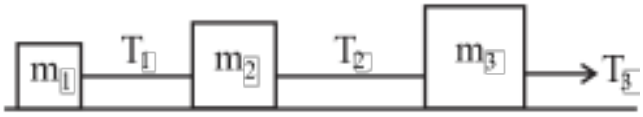

(1) 10 N

(2) 30 N

(3) 50 N

(4) 40 N

39) Three blocks are connected by strings as shown in fig. below, and are pulled by a force $T_3 = 60$ N. If $m_1 = 3$ kg, $m_2 = 4$ kg and $m_3 = 5$ kg. Calculate the tension $T_2$



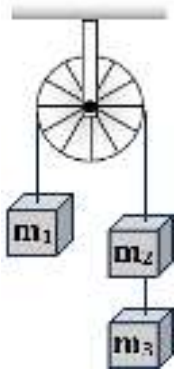

(1) 30 N

(2) 35 N

(3) 20 N

(4) 15 N

40) Three equal masses $m_1$, $m_2$ and $m_3$ of mass 2 kg each are hanging on a string passing over a fixed frictionless pulley as shown in the figure. The tension in the string connecting masses $m_2$ and $m_3$ is



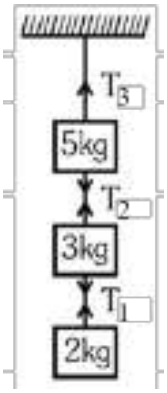

(1) Zero

(2) 13.06 N

(3) 3.3 N

(4) 19.6 N

41) In the given figure determine $T_1 : T_2 : T_3$.



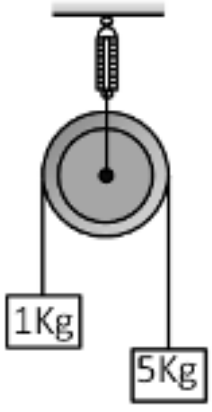

(1) 2 : 3 : 5

(2) 2 : 5 : 10

(3) 10 : 2 : 5

(4) 10 : 5 : 2

42) Find the reading of the spring balance in kg.wt



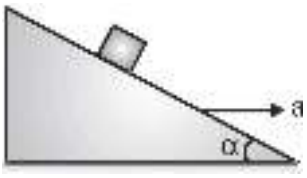

(1) 5/3

(2) 6

(3) 4

(4) 10/3

43) A block is kept on a frictionless inclined surface with angle of inclination 'α'. The incline is given an acceleration a to keep the block stationary. Then a is equal to ?




(1) g
(2) g tan α
(3) g/tan α
(4) g cosec α

44) A machine gun fires a bullet of mass 40 g with a velocity 1200 m $s^{-1}$. The man holding it can exert a maximum force of 144 N on the gun. How many bullets can he fire per second at the most ?

(1) One
(2) Four
(3) Two
(4) Three

45) A rocket of initial mass 1500 kg. ejects gas at a constant rate of 10kg/s. with a relative speed of 5 km/s. What is the acceleration of the rocket 50 seconds after the blast, neglecting gravity :-

(1) 10 $ms^{-2}$
(2) 25 $ms^{-2}$
(3) 50 $ms^{-2}$
(4) 100 $ms^{-2}$

## CHEMISTRY

1) Maximum I.P. will be for which element :-

(1) Na
(2) K
(3) N
(4) O

2) The size of the species, Pb, $Pb^{2+}$, $Pb^{4+}$ decreases as :-

(1) $Pb^{4+} > Pb^{2+} > Pb$
(2) $Pb > Pb^{2+} > Pb^{4+}$
(3) $Pb > Pb^{4+} > Pb^{2+}$
(4) $Pb^{4+} > Pb > Pb^{2+}$

3) The pair for which electron addition in both the atoms are endothermic :-

(1) N, O
(2) Be, F
(3) N, Ne
(4) B, C

4) Which of the following bond is least polar ?

(1) O-F
(2) P-F
(3) P-H
(4) B-F

5) Which one is having maximum E.N. ?

(1) $Mn^{+2}$
(2) $Mn^{+4}$
(3) $Mn^{+6}$
(4) $Mn^{+7}$

6) Which of the following molecular geometry is not produced by $sp^3d$ hybridisation?
.

(1) Trigonal planar ($NO_3^-$)
(2) Sea-Saw ($SF_4$)
(3) ''T'' shape ($BrF_3$)
(4) Linear ($XeF_2$)

7) In Triple bond, there is sharing of:-

(1) 3 electrons
(2) 4 electrons
(3) 6 electrons
(4) Several electrons

8) $PCl_5$ exists but $NCl_5$ does not because :

(1) nitrogen has no vacant d-orbitals
(2) $NCl_5$ is unstable
(3) nitrogen atom is much smaller than P
(4) nitrogen is highly inert

9) Which of the molecule is T-shaped

(1) $SF_4$

(2) $IF_7$

(3) $ClF_3$

(4) $XeF_2$

10) Which of the following molecule has square planar shape

(1) $XeF_4$

(2) $BrF_5$

(3) $ClF_3$

(4) $SF_4$

11) In which following set of compound/ion has linear shape

(1) $CH_4$, $NH_4^+$, $BH_4^-$

(2) $CO_3^{-2}$, $NO_3^-$, $BF_3$

(3) $NO_2^+$, $CO_2$, $XeF_2$

(4) $BeCl_2$, $BCl_3$, $CH_4$

12) Total no. of vacant orbitals in valence shell of sulphur when it undergoes formation of $SF_4$:-

(1) 5

(2) 4

(3) 3

(4) 2

13) Which pair have maximum gap of $Z_{eff}$ ?

(1) Li, Be

(2) B, C

(3) C, F

(4) C, N

14)

| $IE_1$ | $IE_2$ | $IE_3$ | $IE_4$ |
|---|---|---|---|
| 165 | 195 | 556 | 595 |

(in KCal)

Identify the $e^-$ configuration of that element?

(1) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$

(2) $1s^2\ 2s^1$

(3) $1s^2\ 2s^2\ 2p^2$

(4) $1s^2\ 2s^2\ 2p^6\ 3s^2$

15) Which of the following represents correct order of electron affinity ?

(1) Cl > F > S > O

(2) F > O > S > Cl

(3) F > Cl > S > O

(4) Cl > S > O > F

16) Which of the following element has the different value of electronegativity than other three:

(1) H

(2) S

(3) C

(4) I

17)

Which of the following overlaps is incorrect.
(assuming z-axis as inter nuclear axis) ?

(a) $2P_y + 2P_y \rightarrow \pi$ bond
(b) $2P_x + 2P_x \rightarrow \sigma$ bond
(c) $2P_z + 2P_z \rightarrow \pi$ bond
(d) $s + P_z \rightarrow \pi$ bond
(e) $s + P_z \rightarrow \sigma$ bond

(1) a, b, c

(2) b, c, d

(3) a, c, d

(4) c, d, e

18) At equilibrium :-
(a) Rate of forward reaction should be equal to rate of backward reaction
(b) Concentration of reactant and product should be equal
(c) Concentration of reactant and product should be constant
(d) Concentration of reactant and product can never be equal
Correct sequence is (true/false)

(1) TTFT

(2) TFFT

(3) TFTF

(4) TTTF

19) A two litre flask contains 1.4 g nitrogen and 1.0 g hydrogen. The ratio of active mass of nitrogen and hydrogen would be -

(1) 1 : 3

(2) 1 : 5

(3) 1.4 : 1

(4) 1 : 10

20) For the equilibrium,

$MgCO_{3(s)} \rightleftharpoons MgO_{(s)} + CO_{2(g)}$

Which of the following expressions is correct ?

(1) $K_P = P_{CO_2}$

(2) $K_P = \frac{[MgO][CO_2]}{[MgCO_3]}$

(3) $K_P = \frac{P_{MgO} \cdot P_{CO_2}}{P_{MgCO_3}}$

(4) $K_P = P_{MgO} \cdot P_{CO_2}$

21) For which of the following reaction $K_p = K_c$ ?

(1) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(2) $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$

(3) $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$

(4) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

22) $\log \frac{K_p}{K_c} + \log RT = 0$ is true relationship for the following reaction :-

(1) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(2) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

(3) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(4) (2) and (3) both

23) In which of the following reaction, which one has the least stable product :-

(1) $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $K_1 = 2.3 \times 10^{-2}$

(2) $N_2 + O_2 \rightleftharpoons 2NO$, $K_2 = 2 \times 10^2$

(3) $H_2 + I_2 \rightleftharpoons 2HI$, $K_3 = 294$

(4) $XeO + \frac{1}{2}O_2 + F_2 \rightarrow XeO_2F_2$, $K_4 = 1.4 \times 10^{-3}$

24) If the value of an equilibrium constant for a particular reaction is $10^{-14}$, then at equilibrium the system will contain :-

(1) Mostly reactants

(2) Mostly products

(3) Similar amounts of reactants and products

(4) All reactants

25) The equilibrium constants for the reaction, $Br_2(g) \rightleftharpoons 2Br(g)$, at 500 K and 700 K are $1 \times 10^{-10}$ and $1 \times 10^{-5}$ respectively. The reaction is :

(1) Endothermic

(2) Exothermic

(3) Fast

(4) Slow

26) At 527°C, the reaction given below has $K_C = 4$

$$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$$

What is the $K_P$ for the following reaction ?

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

(1) $16 \times (800R)^2$

(2) $\left(\frac{800R}{4}\right)^{-2}$

(3) $\left(\frac{1}{4 \times 800R}\right)^2$

(4) None of these

27) **Statement-I :** For the physical equilibrium $H_2O(s) \rightleftharpoons H_2O(\square)$ on increasing temperature and increasing pressure more water will form.
**Statement-II :** Since forward reaction is endothermic in nature and volume of water is greater than that of volume of ice.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

28) $2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}$
Initially 8 moles of HI were taken. If degree of dissociation of HI is 0.25 then calculate moles of $I_2$ at equilibrium.

(1) 6 moles

(2) 2 moles

(3) 1 mole

(4) 5 mole

29) The hydrogen ion concentration in a given solution is $6 \times 10^{-4}$ M. Its pH will be :-

(1) 6

(2) 3.22

(3) 4

(4) 2

30) Ostwald dilution law is applicable for the ionization of :-

(1) $HNO_3$

(2) $H_2SO_4$

(3) NaOH

(4) HCN

31) Degree of ionization of 0.1N $CH_3COOH$ whose $K_a = 10^{-5}$ is:

(1) $10^{-5}$

(2) $10^{-4}$

(3) $10^{-3}$

(4) $10^{-2}$

32) Which one of the following statements is incorrect?

(1) The degree of ionization of a weak electrolyte increase with dilution.

(2) Strong electrolytes are ionized completely at all concentrations.

(3) Addition of $NH_4Cl$ to $NH_4OH$ increases the ionization of the latter.

(4) Increase of temperature increases the ionization.

33) Which of the following base is weakest ?

(1) AOH ($pK_b = 8.9$)

(2) BOH ($pK_b = 11.4$)

(3) COH ($pK_b = 12.4$)

(4) DOH ($pK_b = 6.9$)

34) At a certain temperature the value of $pK_w$ is 13.4 and the measured pH of solution is 7. The solution is:-

(1) Acidic

(2) Basic

(3) Neutral

(4) Unpredictable

35) For pure water :-

(1) pH increases while pOH decreases with rise in temperature

(2) pH decreases while pOH increases with rise in temperature

(3) both pH and pOH decreases with rise in temperature

(4) both pH and pOH increases with rise in temperature

36) Ionic product of water will increase:-

(1) Pressure is increased

(2) $H^+$ ions are added

(3) $OH^-$ ions are added

(4) Temperature is increased

37) **Statement-I :** Degree of ionisation of $CH_3COOH$ is more in normal water ($H_2O$) in comparison of Heavy water ($D_2O$).
**Statement-II :** Dielectric constant of normal water ($H_2O$) is greater than heavy water ($D_2O$), So polar nature of normal water is greater than heavy water.

(1) Both statement-I and statement-II are true and statement-II is correct explanation of statement-I
(2) Both statement-II and statement-II are true and statement-II is not correct explanation of statement-I
(3) Statement-I is incorrect & statement-II is correct
(4) Statement-I is correct and statement-II is incorrect

38) Which will represent common ion effect ?

(1) $HClO_4$ and $KClO_4$
(2) $H_2SO_4$ and KOH
(3) $CH_3COOH$ and HCl
(4) Both (1) and (3)

39) At 90°C, pure water has $[H_3O]^+ = 10^{-6}$ mol/L. Value of $K_w$ at 90°C

(1) $10^{-14}$
(2) $10^{-8}$
(3) $10^{-6}$
(4) $10^{-12}$

40) pH of water is 7.0 at 25°C. If water is heated to 70°C, the

(1) pH will decreases and solution becomes acidic
(2) pH will increase
(3) pH will remain constant as 7
(4) pH will decrease but solution will be neutral

41) For a solution in which 0.2g NaOH is dissolved in 100 mL, the value of pH will be :-

(1) 10.699
(2) 11.699
(3) 12.699
(4) 13.699

42) The pH of a $10^{-7}$ M solution of HCl in water is:-

(1) 8
(2) –8
(3) between 7 and 8

(4) between 6 and 7

43) 200 mL of $\frac{M}{50}$ HCl are mixed with 100 ml of $\frac{M}{100}$ HCl the pH of resulting solution would be -

(1) 2.22
(2) 1.78
(3) 2
(4) Zero

44) The pH of a solution obtained by mixing 50 mL of 0.4 N HCl and 50 mL of 0.2 N NaOH is:

(1) 13
(2) 12
(3) 1.0
(4) 2.0

45)

Consider the following statements
(a) The pH of a mixture containing 400 mL of 0.1 M $H_2SO_4$ and 400 mL of 0.1 M NaOH will be approximately 1.3
(b) lonic product of water is temperature dependent.
(c) The degree of ionisation of a weak acid increases on dilution.
(d) The Le Chatelier's principle is not applicable to common-ion effect.
The correct statements are:

(1) (a) and (b)
(2) (a), (b) and (c)
(3) (a), (b) and (d)
(4) (b) and (c)

## BIOLOGY

1) Which one of the following statements is wrong?

(1) Algae increase the level of dissolved oxygen in the immediate environment
(2) Algin is obtained from red algae and carrageen from brown algae
(3) Agar-agar is obtained from *Gelidium* and *Gracilaria*
(4) Laminaria and Sargassum are used as food

2) Zygotic meiosis occurs in :-

(1) *Pinus*
(2) *Marchantia*
(3) *Chlamydomonas*

(4) *Dryopteris*

3) The gametophyte is not an independent, free living generation in :-

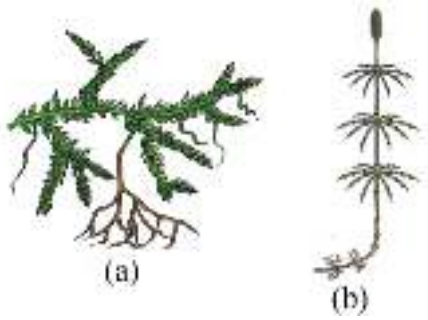
(1) *Polytrichum*
(2) *Adiantum*
(3) *Marchantia*
(4) *Cycas*

4) Peat is obtained from :-

(1) *Riccia*
(2) *Marchantia*
(3) *Sphagnum*
(4) *Funaria*

5) Antherozoids of bryophytes are :-

(1) Biflagellated spores
(2) Biflagellated gametes
(3) Multiflagellated spore
(4) Female gamete

6) 


Identify the plants given in the figure (a) & (b)

(1) *Selaginella* and *Pteris*
(2) *Equisetum* and *Salvinia*
(3) *Selaginella* and *Equisetum*
(4) *Salvinia* and *Azolla*

7) *Selaginella* & *Salvinia* are ____ pteridophyte :-

(1) Homosporous
(2) Heterosporous
(3) Plolyploid
(4) Aquatic

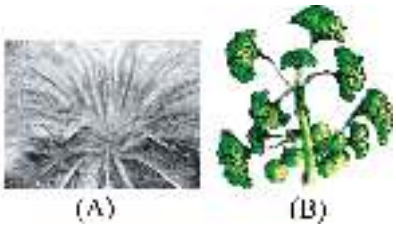
8) Which of the following statement is incorrect about Gymnosperm ?

(1) Pollination takes place by air in gymnosperms.

(2) They are heterosporous

(3) Before and after fertilization, ovules are not enclosed by any ovary wall and remain exposed.

(4) All Gymnosperm are dioecious without any exception.

9) 

Identify the given figures :

(1) A - *Cycas*, B - *Ginkgo*

(2) A - *Cycas*, B - *Pinus*

(3) A - *Ginkgo*, B - *Cycas*

(4) A - *Ginkgo*, B - *Ginkgo*

10) Which statement is correct for pteridophyta ?

(1) Gametophyte is autotrophic, while sporophyte is parasitic.

(2) Sporophyte is semiparasite and gametophyte is autotrophic.

(3) Both gametophyte and sporophyte are usually autotrophic.

(4) Only sporophyte is autotrophic, while gametophyte is parasite.

11) **Assertion :** Spreading of living pteridophytes is limited and restricted to narrow geographical region.
**Reason :** Pteridophytes required water for fertilization.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

12) Arrange the following events of life cycle of bryophytes in correct sequence :-
I. Germination of spore
II. Gametes formation
III. Formation of gametophyte
IV. Fertilization
V. Embryo formation

(1) I,III,IV,V,II

(2) II,IV,I,V,III

(3) I,III,II,IV,V

(4) V,I,II,III,IV

13) Which is not true about agar :-

(1) It is obtained from red algae

(2) It is used to grow microbes

(3) It is used to make ice-cream and jellies

(4) It is used as food supplement even by space travellers

14) The given table shows classes of algae and their main characteristics. Some characteristics in this table mentioned as A, B, C and D, Observe the table and choose the option in which A, B, C and D are correctly identified :-

| Classes | Chlorophyceae | Phaeophyceae | Rhodophyceae |
|---|---|---|---|
| Major Pigments | Chl a, b | Chl a, c | **A** |
| Stored Food | Starch | **B** | Floridean starch |
| Cell Wall | **C** | Cellulose | Cellulose, pectin, polysulphate ester |
| Flagellar Number | 2-8 | 2 | **D** |

| | A | B | C | D |
|---|---|---|---|---|
| (1) | Chl a, e | Fat & oil | Peptidoglycan | Many |
| (2) | Chl a, d | Mannitol, Laminarin | Cellulose | Absent |
| (3) | Chl a, d | Paramylum | Lipoprotein | One |
| (4) | Chl a, c | Glycogen | Cellulose | Absent |

(1) 1

(2) 2

(3) 3

(4) 4

15) Read the following paragraph and fill in the blanks with approprirte words/ terms.
In pteridophytes the main plant body is a ____(A)____which bears sporangia that are subtended by leaf like appendages called ___(B)____ The sporangia produce spores by ___(C)____ and after germination of spores, they give rise to photosynthetic gametophytes called ____(D)____.
Choose the correct option :-

| | A | B | C | D |
|---|---|---|---|---|
| (1) | Sporophyte | sporangia | meiosis | Protonema |
| (2) | Gametophyte | Sporophylls | mitosis | Protonema |
| (3) | Sporophyte | Sporophylls | meiosis | Prothallus |

| | | | | |
|---|---|---|---|---|
| (4) | Gametophyte | Sporophylls | mitosis | Prothallus |

(1) 1

(2) 2

(3) 3

(4) 4

16) Consider the following four statements A, B, C and D and select the right option for two correct statements :-
(A) Diplontic life cycle is found in all seed bearing plants
(B) In *Pinus*, both male cones and females cones are borne on same tree.
(C) Male gametes of *Pinus* are biflagellated
(D) In *Pinus*, coralloid roots are present
The correct statements are :-

(1) A and B

(2) B and C

(3) C and D

(4) A and C

17) The primary acceptor of $CO_2$ in $C_4$ plants has :–

(1) 2 carbons

(2) 6 carbons

(3) 3 carbons

(4) 4 carbons

18) Glycolate cycle/$C_2$ cycle occurs in which of the following organelle(s) :

(1) Glyoxysome

(2) Sphaerosome

(3) Peroxisome, lysosome, glyoxysome

(4) Peroxisome, mitochondria, chloroplast

19) In $C_3$ cycle, Fixation of 1 $CO_2$ requires :–

(1) 6 $NADPH_2$ & 3 ATP

(2) 2 $NADPH_2$ & 3 ATP

(3) 4 $NADPH_2$ & 3 ATP

(4) 5 $NADPH_2$ & 3 ATP

20) The biosynthetic phase of photosynthesis depends on the products of light reaction i.e. :

(1) ATP & $O_2$

(2) NADPH & $CO_2$

(3) NADPH & $O_2$

(4) ATP & NADPH

21) According to chemiosmotic hypothesis, ATP synthase form ATP when protons ($H^+$) :-

(1) Move from stroma to thylakoid lumen by simple diffusion.
(2) Move from stroma to thylakoid lumen by facilitated diffusion.
(3) Move from thylakoid lumen to stroma by active transport.
(4) Move from thylakoid lumen to stroma by facilitated diffusion.

22) Which one is the most abundent enzyme on earth ?

(1) RuBisCO
(2) PEPcase
(3) Hyxokinase
(4) Pyruvate acid kinase

23) Photosynthesis is :

(1) Catabolic and Exergonic process
(2) Anabolic and Exergonic process
(3) Anabolic and Redox process
(4) Catabolic and Redox process

24) Match the column :-

| **Column-I** | | **Column-II** | |
|---|---|---|---|
| (a) | Jan Ingenhousz | (i) | First action spectrum of photosynthesis |
| (b) | Joseph priestley | (ii) | Essentiality of sunlight to plants |
| (c) | T. W. Engelmann | (iii) | Discovered oxygen |

(1) a-(i), b-(ii), c-(iii)
(2) a-(ii), b-(i), c-(iii)
(3) a-(iii), b-(ii), c-(i)
(4) a-(ii), b-(iii), c-(i)

25) Moll's half leaf experiment proves that :-

(1) Chlorophyll is essential for photosynthesis
(2) $CO_2$ is essential for photosynthesis
(3) $O_2$ released in photosynthesis comes from $H_2O$
(4) Both (1) and (2)

26)

Match the following.

| | Pigment | | Color (in chromatogram) |
|---|---|---|---|
| A | chl-a | i | Bright or blue green |
| B | chl-b | ii | yellow |
| C | xanthophyll | iii | yellow green |
| D | carotenoids | iv | yellow to yellow orange |

(1) A - i, B - ii, C - iii, D - iv

(2) A - i, B - iii, C - ii, D - iv

(3) A - ii, B - i, C - iii, D - iv

(4) A - iii, B - ii, C - i, D - iv

27) Which of the following is **wrongly** matched ?

(1) Sorghum — Kranz anatomy

(2) Melvin Calvin — Used radioactive- $C^{14}$

(3) Photorespiration — $C_3$ plants

(4) PS-II — $P_{700}$

28)

Saturation of $CO_2$ concentration for maize and wheat respectively are :-

(1) 450 $\mu L^{-1}$ and 360 $\mu L^{-1}$

(2) 360 $\mu L^{-1}$ and 450 $\mu L^{-1}$

(3) 360 $\mu L^{-1}$ and 360 $\mu L^{-1}$

(4) 450 $\mu L^{-1}$ and 450 $\mu L^{-1}$

29) All is related to the $C_4$ plants except :

(1) Kranz Anatomy

(2) Double $CO_2$ fixation

(3) Sugar synthesis in mesophyll cells

(4) Adapted to dry tropical conditions

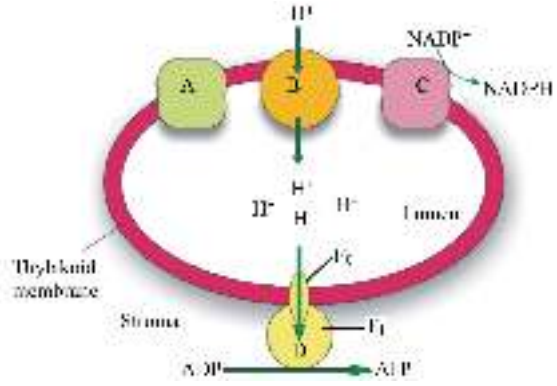
30) Which of the following compound is not related to $C_3$ cycle :-

(1) $NADPH + H^+$

(2) PGA

(3) Pyruvate

(4) RuBP

31) 


Choose the correct option with respect to A,B,C & D given in the diagram :-

| | A | B | C | D |
|---|---|---|---|---|
| 1 | PS II | Cytochrome b & f | ATP synthase | PS I |
| 2 | PS II | Cytochrome b & f | PS I | ATP synthase |
| 3 | PS I | PS II | ATP synthase | Cytochrome b & f |
| 4 | PS I | Cytochrome b & f | ATP synthase | PS II |

(1) 1

(2) 2

(3) 3

(4) 4

32) Find out the correct match from following table :-

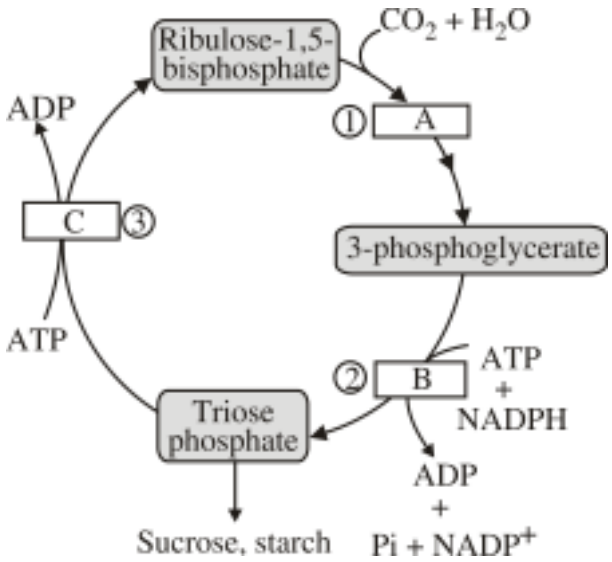
| | **Characteristics** | **$C_3$ plants** | **$C_4$ plants** |
|---|---|---|---|
| I | Primary $CO_2$ acceptor | RuBP | RuBP |
| II | Primary $CO_2$ fixation product during $C_3$ cycle | PGA | PGA |
| III | No. of carbons in the primary $CO_2$ acceptor | 3 | 5 |
| IV | Initial carboxylation reaction occurs | Mesophyll cells | Bundle sheath cells |

(1) I, III, IV

(2) II only

(3) IV only

(4) II, III, IV




33) In given figure, A, B and C are respectively-

(1) A-Carboxylation; B-Reduction; C-Transport

(2) A-Reduction; B-Transport; C-Carboxylation

(3) A-Carboxylation; B-Reduction; C-Regeneration

(4) A-Regeneration; B-Reduction; C-Carboxylation

34) Read the following four statements (A-D) and select the correct option ones ? (A) The medulla is divided into a few conical masses projecting into the calyces.
(B) The cortex extends in between the medullary pyramids as renal columns called columns of Bertini.
(C) Kidneys are blackish brown, bean shaped structures situated between the levels of last thoracic and third lumbar vertebra close to the ventral inner wall of the abdominal cavity.
(D) Each kidney of an adult human measures 10-12 cm in length, 5-7 cm in width,2-3 cm in thickness with an average weight of 120-170 gm.

(1) Statements A, B & C

(2) Statements B, C & D

(3) Statement A, B & D

(4) Statement B and C

35)

Which one is the primary function of sweat glands in human?

(1) To facilitate a cooling effect on the body surface.

(2) To help in removal of some of the wastes like NaCl, urea, Lactic acid etc.

(3) It helps in osmoregulation.

(4) It maintains the NaCl in our body fluid.

36) Dialysing unit (Artificial kidney) contains a fluid which is almost same as plasma except that it has -

(1) High nutrients

(2) High nitrogenous wastes.

(3) No nitrogenous wastes

(4) High NaCl & urea

37) Match the abnormal conditions given in column-A, with their explanations given in column-B and select the correct option ?

| | **Column-A** | | **Column-B** |
|---|---|---|---|
| (A) | Uremia | (i) | Accumulation of urea in urine |
| (B) | Renal calculi | (ii) | Presence of Glucose in urine |
| (C) | Glomerulone-phritis | (iii) | Mass of crystallised salt with in kidney |
| (D) | Glycosuria | (iv) | Inflammation in glomeruli |
| | | (v) | Accumulation of urea in blood |

(1) A-ii, B-iii, C-iv, D-i

(2) A-i, B-ii, C-iii, D-iv

(3) A-v, B-iii, C-iv, D-ii

(4) A-v, B-iii, C-iv, D-i

38) We have to observe that glucose and ketone bodies pass through urine in a person. Such condition indicates the sign of :-

(1) Diabetes insipidus

(2) Glomerulonephritis

(3) Diabetes mellitus

(4) Renal failure.

39) Which of the following statements is correct ?

(1) ADH :- Conversion of angiotensinogen in blood to angiotensin.

(2) Aldosterone :- Reabsorption of $Na^+$ and water.

(3) ANF :- Enhance $Na^+$ reabsorption.

(4) Renin :- Causes vasodilation.

40) Substances like amino acid and glucose in the filtrate are reabsorbed .....(A)..... in the tubular epithelial cells whereas the nitrogenous wastes product are reabsorbed by .....(B)..... What are the A & B and select the correct option ?

(1) A = Actively; B = Passively

(2) A = Passively; B = Actively

(3) A = Actively; B = Actively

(4) A = Passively; B = Passively

41) Identify the **correct** combination :-

(1) Green glands – crustacean – uricotelic

(2) Malpighian tubule – insect – uricotelic

(3) Nephridia – annelida – uricotelic

(4) Flame cells – Aschelminthes - uricotelic

42)

How many statements are correct ?
(a) vasa recta is absent or highly reduced in cortical nephrons
(b) ammonia is less toxic as compared to urea and requires large amount of water for its elimination
(c) GFR in a healthy individual is approximately 125 ml / minute
(d) PCT also help to maintain the pH and ionic balance of the body fluid

(1) one

(2) two

(3) three

(4) four

43) Glucose and 75% water is absorbed in :-

(1) proximal convoluted tubule

(2) distal convoluted tubule

(3) collecting duct

(4) loop of Henle

44) **Assertion :** Kidney maintains the osmotic concentration of the blood.
**Reason** : Kidney eliminates either hypotonic or hypertonic urine according to the need of the body.

(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) If Assertion is true but Reason is false.

(4) If both Assertion and Reason are false.

45) The terms “Ammonotelic”, “Ureotelic” and “Uricotelic” are used to describe-

(1) Modes of excretory system development

(2) The actions of hormones on the excretory systems

(3) The types of nitrogenous waste produced by various classes of vertebrates

(4) Modification of kidney tubules to enhance excretion

46) Which of the following statements is false?
I. Outer cortex and inner medulla are the two zones in kidney
II. Medulla is divided into about 8 to 12 medullry pyramids
III. Pyramid projects into calyx
IV. Inwards extension of cortex between the pyramids is called renal column of Bertini

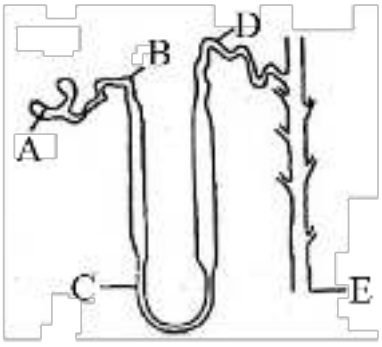
(1) I and IV

(2) II and IV

(3) IV

(4) None

47) Use following diagram to complete the statements about the human nephron-




I. The composition of the filtrate would be most like plasma in the tubule.
II. The urine would be most concentrated in the collecting duct.
III. Most of the glomerular filtrate is reabsorbed into peritubular capillary.
IV. Conducting of urine to pelvis of the kidney from the structure.
V. Most water is reabsorbed by the structure.

(1) I-A, II-C, III-B, IV-E, V-D

(2) I-A, II-E, III-B, IV-C, V-D

(3) I-A, II-B, III-E, IV-C, V-D

(4) I-A, II-E, III-B, IV-E, V-B

48) Which of the following sequence is correct?

(1) An increase in body fluid volume → switch off the osmoreceptors → suppresses the ADH release

(2) ADH-Constricting effect on blood vessel → B.P. high → Glomerular blood flow more → GFR more

(3) Angiotensinogen → Angiotensin I → Angiotensin-II → Adrenal cortex → Aldosterone

(4) All

49)

Following are the steps of dialysis-
A. Blood is passed into a vein.
B. Blood is mixed with heparin.
C. Blood is mixed with anti-heparin.
D. Blood is drained from convenient artery.
E. Blood is passed through a coiled and porous cellophane tube bathing in dialysis fluid.
F. Removal of nitrogenous wastes from blood.
The correct sequence of steps is-

(1) A → B → C → D → E → F

(2) F → C → E → B → A → D

(3) D → B → E → F → C → A

(4) D → C → E → F → B → A

50) A man has taken large amount of protein in his diet. He will excrete more of :-

(1) Urea
(2) Uric acid
(3) Sugar
(4) Salt and water

51)

Match the following :-

| Column-I | | Column-II | |
|---|---|---|---|
| A. | PCT | (i) | Selective secretion of $H^+$ and $K^+$ ions. |
| B. | DCT | (ii) | 70-80 percent of electrolytes and water reabsorbed |
| C. | Descending limb of LOH | (iii) | Allow passage of small amount of urea. |
| D. | Ascending limb of LOH | (iv) | Permeable to water |
| E. | Collecting duct | (v) | Permeable to salts |

(1) A-(i), B-(ii), C-(iii), D-(iv), E-(v)
(2) A-(i), B-(iii), C-(iv), D-(ii), E-(v)
(3) A-(ii), B-(i), C-(iv), D-(v), E-(iii)
(4) A-(ii), B-(i), C-(v), D-(iv), E-(iii)

52) Read the following points :-
(i) Increases GFR
(ii) Increases B.P.
(iii) Decreases blood volume
(iv) Increases aldosterone secretion
(v) Vasoconstrictor
(vi) Increases $Na^{\oplus}$ excretion
How many points are correct about Angiotensin II?

(1) Two
(2) Three
(3) Four
(4) Five

53) Which one is reabsorbed actively in the nephron?

(1) Glucose and amino acid only
(2) Glucose, amino acid and $Na^+$

(3) Glucose and $Na^+$ only

(4) Glucose only

54) Which of the following defines net filteration pressure ?

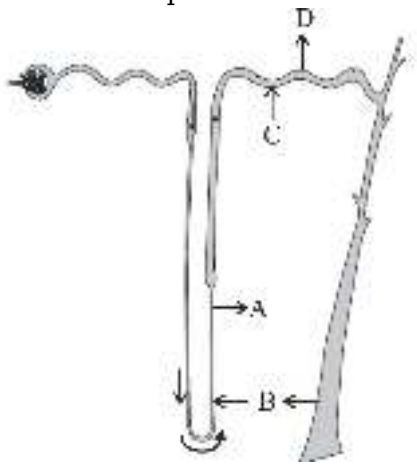
(1) BCOP – (GHP + CHP)

(2) (BCOP + GHP) – CHP

(3) GHP – (BCOP + CHP)

(4) (GHP – CHP) + BCOP

55) Reabsorption and secretion of major substances at different parts of the nephron are shown in




the given figure. In this figure A, B, C and D are respectively:-

(1) $HCO_3^-$, nutrients, urea, NaCl

(2) NaCl, urea, $K^+$, $HCO_3^-$

(3) NaCl, urea, $HCO_3^-$, ammonia

(4) $H_2O$, urea, NaCl, $HCO_3^-$

56) First step in urine formation is–

(1) Ultrafiltration

(2) Tubular secretion

(3) Selective secretion

(4) Tubular reabsorption

57) ANF acts as a check on ?

(1) ADH mechanism

(2) Myogenic mechanism

(3) RAAS mechanism

(4) Chloride shift

58) Osmolarity of Inner medulla is approximately?

(1) 300 mOsmol/L

(2) 600 mOsmol/L

(3) 1200 mOsmol/L

(4) 5000 mOsmol/L



59) Identify following figure and choose **correct** option

(1) DCT

(2) PCT

(3) Loop of henle

(4) Malpighian body

60) Major site of reabsorption and selective secretion in nephron is :-

(1) PCT

(2) DCT

(3) Loop of Henle

(4) PCT and loop of Henle

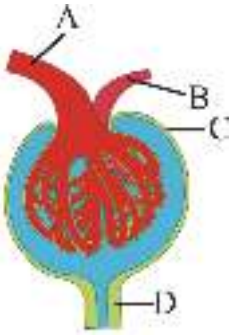
61) Which of the following play a significant role in the maintenance of medullary interstitial osmolarity gradient?

(1) Tubular secretion

(2) Reabsorption of water by action of ADH

(3) Counter current mechanism

(4) Reabsorption of $Na^+$ by action of aldosterone




62) Identify A-D in the following structure and choose the correct option:-

(1) A–Afferent arteriole, B–Efferent arteriole,
C–Bowman's capsule,
D–Proximal convoluted tubule

(2) A–Efferent arteriole, B–Afferent arteriole,
C–Bowman's capsule, D–DCT

(3) A–Efferent arteriole, B–Efferent arteriole,
C–Bowman's capsule, D–DCT

(4) A–Efferent arteriole, B–Afferent arteriole,
C–Glomerulus, D–DCT

63)

In the given how many are the secretions of sebaceous glands ?

Sterols, Lactic acid, Hydrocarbons, Waxes, Urea

(1) 5

(2) 4

(3) 3

(4) 2

64) Which of the following structure have stretch receptors for micturition?

(1) Urinary bladder

(2) Ureters

(3) Renal pelvis

(4) Column of Bertini

65) Which of the following is produced by kidney ?

(1) Renin

(2) Erythropoietin

(3) Calciferol

(4) Both 1 & 2

66) The part of kidney gateway for ureter, nerves and blood vessels is :-

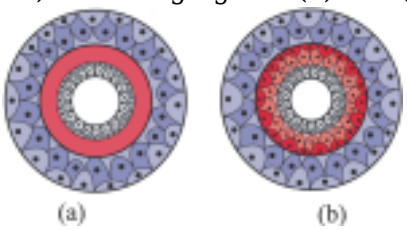
(1) Hilus/Hilum

(2) Renal pore

(3) Minor calyx

(4) Major calyx

67) Following figures (a) and (b) represented the condition animals which are respectively :-




(1) (a)-Diploblastic, (b)-Diploblastic

(2) (a)-Triploblastic, (b)-Triploblastic

(3) (a)-Diploblastic, (b)-Triploblastic

(4) (a)-Triploblastic, (b)-Diploblastic

68) Choose the incorrect statement about ctenophores :-

(1) Commonly known as sea walnut or comb jellies

(2) Exclusively marine animals

(3) Bioluminescence property is well marked

(4) Called flatworms

69) Cells present in sponges which make the lining of spongocoel and also known as collar cells are ?

(1) Spongioblasts

(2) Archaeocytes

(3) Choanocytes

(4) Solenocytes

70) Animals with a pseudocoelom, often referred to as roundworms, belong to which phylum?

(1) Platyhelminthes

(2) Aschelminthes

(3) Annelida

(4) Coelenterata

71) Consider the following statements regarding symmetry and coelom :
I. All triploblastic animals possess a true coelom.
II. Radial symmetry is characteristic of both Cnidarians and adult Echinoderms.
III. Pseudocoelomates are bilaterally symmetrical.
Which of the statements are correct?

(1) I and II only

(2) II and III only

(3) I and III only

(4) I, II and III

72) Match the following subphylum of phylum chordata with their characters and examples:

| **Subphylum** | **Characters** | **Examples** |
|---|---|---|
| (A) Urochordata | (a) Notochord replaced by vertebral column | (i) Amphioxus |
| (B) Cephalochordata | (b) Notochord is present only in larval tail | (ii) Ascidia |
| (C) Vertebrata | (c) Notochord present from head to tail region and is persistent throughout their life | (iii) Fish |

(1) A-b-ii ; B-c-i ; C-a-iii

(2) A-a-ii ; B-c-i ; C-b-ii

(3) A-c-i ; B-a-iii ; C-b-ii

(4) A-b-iii ; B-a-i ; C-c-ii

73) Radula of molluscs related to :-

(1) Digestion
(2) Respiration
(3) Circulation
(4) Excretion

74) $NH_3$ is

(1) Less toxic
(2) Toxic
(3) Highly toxic
(4) None

75) The statement not applicable for sponges is :-

(1) They have calcareous spicules
(2) They all have radial symmetry
(3) Cellular level of organisation
(4) Sexual reproduction occurs by gametes.

76) Water enter through (In porifera) :-

(1) Osculum
(2) Ostia
(3) Spicules
(4) Spongocoel

77) Read the following statements and choose the correct option.
(A) In Mollusca, the space between the hump and the mantle is called the mantle cavity.
(B) In Echinodermates, digestive system is complete with mouth on the lower (ventral) side.
(C) Hemichordata was earlier considered as a class under phylum Chordata.
(D) In Mollusca, A hard and spongy layer of skin forms a mantle over the visceral hump.

(1) A-T, B-T, C-F, D-F
(2) A-F, B-T, C-T, D-F
(3) A-T, B-F, C-T, D-T
(4) A-F, B-T, C-F, D-T

78) Arthropoda are most successful terrestrial invaders because :-

(1) Compound eye, antenna, wings is present
(2) Chitinous exoskeleton, wings, compound eye is present
(3) Chitinous exoskeleton, jointed appendages

(4) Jointed appendages, wings, antenna is present

79) Match the column-I with column-II :

| Column-I | | Column-II | |
|---|---|---|---|
| (a) | Porifera | (i) | Nephredia |
| (b) | Coelenterata | (ii) | Cnidoblast |
| (c) | Platyhelminthes | (iii) | Flame cell |
| (d) | Annelida | (iv) | Choanocyte cell |

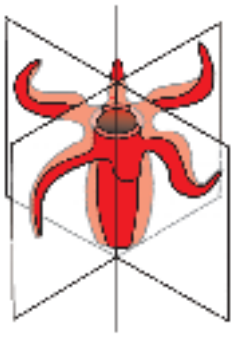
(1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(2) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(3) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

(4) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

80) The given below figure is showing to which type of symmetry ?



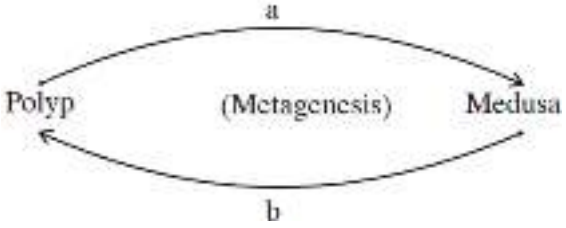
(1) Bilateral

(2) Radial

(3) Asymmetry

(4) None of the above

81) Recognise the figure and find out the correct option about a and b :-

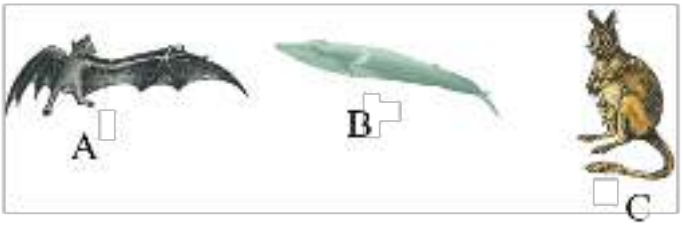



(1) a-sexually, b-asexually

(2) a-asexually, b-sexually

(3) a-asexually, b-asexually

(4) a-sexually, b-sexually

82)



A, B, and C animals are respectively :-

(1) Pteropus, Balaenoptera, Macropus
(2) Macropus, Balaenoptera, Pteropus
(3) Balaenoptera, Pteropus, Macropus
(4) Balaenopter, Macropus, Pteropus

83) Character found only in mammals:-

(1) Homeothermy
(2) Viviparity
(3) Dicondylic skull
(4) Muscular diaphragm

84) In which of the following tympanum is absent:-

(1) Birds
(2) Frogs
(3) Lizards
(4) Snakes

85) **Assertion :** Coelenterates and ctenophores are consider as radiata.
**Reason :** Any plane passing through central axis of body divides organism into two identical halves.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

86) Body is unsegmented covered by a calcareous shell and is divided into head, muscular foot and visceral hump, excretory system include 1 or 2 pairs of kidneys known as Keber's organ. These characters are found in which phylum ?

(1) Mollusca
(2) Echinodermata
(3) Hemichordata
(4) Aschelminthes

87) Which of the following pair of animals are referred as 'Jawless fishes' ?

(1) Magur & flying fish

(2) Lamprey & Eel

(3) Rohu & Dog fish

(4) Hag fish & Lamprey

88)

Following features belongs to which class ?
(a) Body is covered by dry and cornified skin, epidermal scales or scutes
(b) Tympanum represent the ear
(c) Poikilotherms animals

(d) Heart is usually three chambered

(1) Reptilia

(2) Amphibia

(3) Pisces

(4) Aves

89) Which of the following annelid is found in marine water and locomote through parapodia?

(1) Leech

(2) Earthworm

(3) Bonellia

(4) Nereis

90) Parasitic vertebrate is:

(1) Frog

(2) *Taenia*

(3) *Wuchereria*

(4) *Myxine*

ANSWER KEYS

## PHYSICS

| Q. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 1 | 1 | 3 | 4 | 3 | 2 | 1 | 1 | 3 | 2 | 4 | 4 | 2 | 2 | 1 | 4 | 4 | 2 | 3 | 2 |
| Q. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| A. | 3 | 2 | 2 | 3 | 4 | 4 | 2 | 3 | 2 | 2 | 3 | 3 | 1 | 3 | 3 | 1 | 3 | 4 | 2 | 2 |
| Q. | 41 | 42 | 43 | 44 | 45 | | | | | | | | | | | | | | | |
| A. | 2 | 4 | 2 | 4 | 3 | | | | | | | | | | | | | | | |

## CHEMISTRY

| Q. | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 3 | 2 | 3 | 3 | 4 | 1 | 3 | 1 | 3 | 1 | 3 | 2 | 3 | 4 | 1 | 1 | 2 | 3 | 4 | 1 |
| Q. | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |
| A. | 2 | 2 | 4 | 1 | 1 | 3 | 2 | 3 | 2 | 4 | 4 | 3 | 3 | 2 | 3 | 4 | 1 | 3 | 4 | 4 |
| Q. | 86 | 87 | 88 | 89 | 90 | | | | | | | | | | | | | | | |
| A. | 3 | 4 | 2 | 3 | 2 | | | | | | | | | | | | | | | |

## BIOLOGY

| Q. | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 | 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 | 110 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 2 | 3 | 4 | 3 | 2 | 3 | 2 | 4 | 1 | 3 | 1 | 3 | 4 | 2 | 3 | 1 | 3 | 4 | 2 | 4 |
| Q. | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 | 121 | 122 | 123 | 124 | 125 | 126 | 127 | 128 | 129 | 130 |
| A. | 4 | 1 | 3 | 4 | 2 | 2 | 4 | 2 | 3 | 3 | 2 | 2 | 3 | 3 | 1 | 3 | 3 | 3 | 2 | 1 |
| Q. | 131 | 132 | 133 | 134 | 135 | 136 | 137 | 138 | 139 | 140 | 141 | 142 | 143 | 144 | 145 | 146 | 147 | 148 | 149 | 150 |
| A. | 2 | 3 | 1 | 1 | 3 | 4 | 4 | 4 | 3 | 1 | 3 | 3 | 2 | 3 | 2 | 1 | 3 | 3 | 4 | 1 |
| Q. | 151 | 152 | 153 | 154 | 155 | 156 | 157 | 158 | 159 | 160 | 161 | 162 | 163 | 164 | 165 | 166 | 167 | 168 | 169 | 170 |
| A. | 3 | 1 | 3 | 1 | 4 | 1 | 3 | 4 | 3 | 2 | 2 | 1 | 1 | 3 | 2 | 2 | 1 | 3 | 3 | 2 |
| Q. | 171 | 172 | 173 | 174 | 175 | 176 | 177 | 178 | 179 | 180 | | | | | | | | | | |
| A. | 2 | 1 | 4 | 4 | 1 | 1 | 4 | 1 | 4 | 4 | | | | | | | | | | |

SOLUTIONS

PHYSICS

1) They meet at centroid

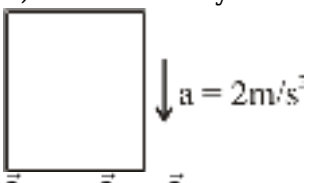



$r = \frac{d/2}{\cos 30^\circ} = \frac{d}{\sqrt{3}}$

$t = \frac{d/\sqrt{3}}{v\cos 30^\circ} = \frac{d/\sqrt{3}}{v\sqrt{3}/2} = \frac{2d}{3v}$

2) When he swims due north, he goes across the river with 30 metre per minute. If he strikes upwards, that is due west of north, his velocity across the river will be less than 30 metres per minute. In this case however, he will reach the opposite bank at some distance downstream.

3) Initial velocity of ball w.r.t. lift = 0




$\vec{a}_{B/L} = \vec{a}_B - \vec{a}_L$

$= -g - (-a)$

$\vec{a}_{B/L} = -10 + 2 = -8\ m/s^2$

$T = \sqrt{\frac{2h}{a_{B/L}}} = \sqrt{\frac{2\times 2}{8}} = \frac{1}{\sqrt{2}}\ sec$

4)

for collision vertical component of velocity of both must be same

$25\sin\theta = 10\sqrt{3}\sin 60$

$\sin\theta = \frac{3}{5} \Rightarrow \theta = 37°$

5)

$v_{rel} = \sqrt{10^2 + 3^2} = \sqrt{109}$ km/hr

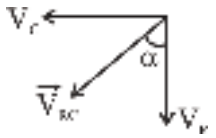
$\vec{v}_{rain}$ and $\vec{v}_{man}$ are perpendicular to each other

6) Since range is max, therefore $\theta = 45°$

Hence, $V_x = V\cos\theta = V\cos 45^\circ = \frac{V}{\sqrt{2}}$

At the highest point, the net velocity of the projectile is $V_x = V\cos45°$

$K.E. = \frac{1}{2}mV_x^2 = \frac{1}{2}m\frac{V^2}{2} = 0.5K$

7) 


$\tan\alpha = \frac{V_C}{V_R} = \frac{40}{20} \times \frac{5}{18} = \frac{5}{9}\ \alpha = \tan^{-1}\left(\frac{5}{9}\right)$

8) $\frac{h_1}{h_2} = \frac{u^2\sin^2 60^\circ/2g}{u^2\sin^2 30^\circ/2g}$

$\frac{h_1}{h_2} = \frac{3}{4} \times \frac{4}{1} = 3:1$

9) Horizontal distance = $u\sqrt{\frac{2H}{g}}$

$R = 150\sqrt{\frac{2 \times 78.4}{9.8}}$

$R = 150 \times 4 = 600$ m

10) Velocity will be perpendicular to acceleration at highest point.

11) speed of man $(v_m) = \frac{d}{90}$

speed of escalator $(v_s) = \frac{d}{60}$

$t = \frac{d}{v_m + v_s} = \frac{d}{\frac{d}{90} + \frac{d}{60}} = \frac{60 \times 90}{150} = 36$ sec

12) $V_A = 36\,\hat{j}$

$V_B = 36\,\hat{i}$

$V_{BA} = V_B - V_A$

$V_{BA} = 36\hat{i} - 36\hat{j}$

$|V_{BA}| = 36\sqrt{2}$ km/hr

$|V_{BA}| = 10\sqrt{2}$ m/s

13) $\frac{H}{R} = \frac{\tan\theta}{4}$

$\theta = 45°$ & $R = 36$ m

$H = 9$ m

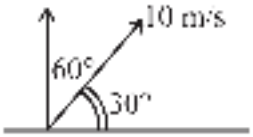
14) $\vec{v} = 20\hat{i} + 40\hat{j}$

$\vec{v} = u\cos\theta\,\hat{i} + u\sin\theta\,\hat{j}$

$T = \frac{2u\sin\theta}{g}$

$T = \frac{2 \times 40}{10}$

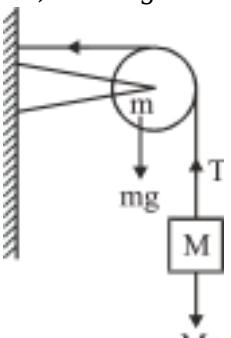
$T = 8s$

15) 


$v = u\cos 30°$

$= \frac{10 \times \sqrt{3}}{2} = 5\sqrt{3}$

16) $T = Mg$



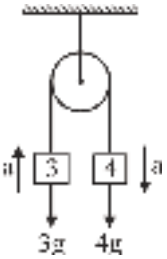

$F_{net} = \left[\sqrt{(M)^2 + (M+m)^2}\right] g$

17) Accleration of system $= \frac{90}{30+2} = 3$ m/s$^2$

Net force actin on the body = ma = (30) (3) = 90 N

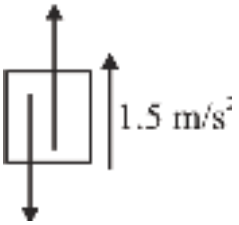
18) $F = \frac{udm}{dt} = m(g+a)$

$\Rightarrow \frac{dm}{dt} = \frac{m(g+a)}{u} = \frac{5000 \times (10+20)}{800} = 187.5$ kg s$^{-1}$

19) 


$$a = \frac{(4-3)g}{(4+3)} = \frac{g}{7} = \frac{9.8}{7} = 1.4 \text{ m/s}$$

20) FBD of 20 kg




N – 20 g = 20 (1.5) ⇒ N = 230 newton

21) Initial momentum of the ball

$$= \frac{150}{1000} \times 12 = 1.8 \text{kg.m/sec}$$

Final momentum of the ball = $\frac{150}{1000} \times 20 = -3.0 \text{ kg m/sec}$

Change in momentum = 4.8 kg m/sec

Average force exerted = Impulse/time = $\frac{4.8}{.01} = 480\text{N}$

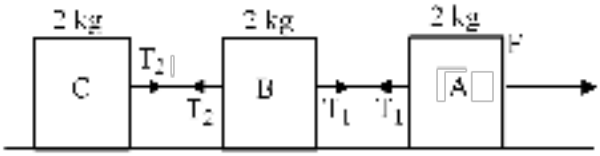
22) Force = mass × acceleration

 F = (m + m + m) × a

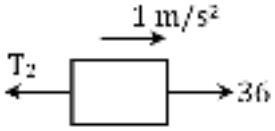
F = 3m × a

$\Rightarrow a = \frac{F}{3m} \Rightarrow a = \frac{10.2}{6} \text{m/s}^2$

 $T_2$ = ma = 2 × $\frac{10.2}{6} = 3.4\text{N}$




23) $a = \frac{36}{1+8+27} = 1 \text{ m/s}^2$




36 – $T_2$ = 27 × 1

$T_2 = 9$ N

24) Assuming "Fig" shows the 5 kg mass ($m_1$) suspended by a string with tension $T_1$ and the 3 kg mass ($m_2$) is suspended below the 5 kg mass by a string with tension $T_2$. The entire system accelerates upwards at a = 2 m/s$^2$.

**1. To find the tension $T_2$ (in the lower string) :**

We consider the forces acting on the lower mass, $m_2$ = 3 kg. The net force is the upward tension $T_2$ minus the downward weight ($m_2g$).

According to Newton's second law ($F_{net} = ma$) :

$T_2 - m_2g = m_2a$

$T_2 = m_2(g + a)$

$T_2 = 3$ kg $\times$ (9.8 m/s$^2$ + 2 m/s$^2$)

$T_2 = 3 \times 11.8$

$T_2 = 35.4$ N

**2. To find the tension $T_1$ (in the upper string) :**

We consider the entire system (both masses) as a single object with a total mass of $M = m_1 + m_2 = 5 + 3 = 8$ kg.

The net force is the upward tension $T_1$ minus the total downward weight (Mg).

$T_1 - (m_1 + m_2)\, g = (m_1 + m_2)a$

$T_1 = (m_1 + m_2)\, (g + a)$

$T_1 = (5 + 3)$ kg $\times$ (9.8 m/s$^2$ + 2 m/s$^2$)

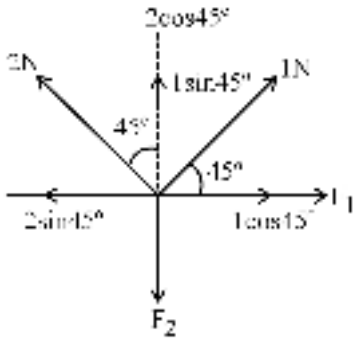
$T_1 = 8 \times 11.8$

$T_1 = 94.4$ N

**Answer :**

The tension are $T_1 = 94.4$ N and $T_2 = 35.4$ N.

25)




$$F_1 + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} \Rightarrow F_1 = \frac{1}{\sqrt{2}}$$

$$F_2 = \frac{2}{\sqrt{2}} + \frac{1}{\sqrt{2}} \Rightarrow F_2 = \frac{3}{\sqrt{2}}$$

26) The apparent weight (N) measured by the scale is given by the formula :

$N = m(g - a)$

Where:

- m = mass = 50 kg
- g = acceleration due to gravity = 10 m/s$^2$
- a = downward acceleration of the life = 9 m/s$^2$

**Calculation:**

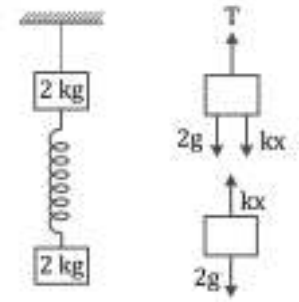
N = 50 × (10 – 9)
N = 50 × 1
N = 50 Newtons
The reading on the weighing scale in kilograms is:

Reading = $\frac{N}{g} = \frac{50}{10} = 5kg$

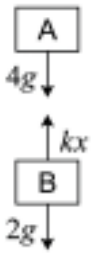
**Answer:** The reading of the weighing scale would be **5 kg**.

27) $a = \frac{10}{5} = 2$ m/s
$T_A$–10 = 2(2) and $T_A$ = 14N
$T_B$ – 10 = 4(2)
$T_B$ = 18 N

28)




kx = 2g
T = 2g + kx = 4g
As soon as the string is cut, T disappears.




 Acceleration ⇒ $\frac{4g}{2}$ = 20 m/s$^2$
⇒ No net force ⇒ 0 m/s$^2$

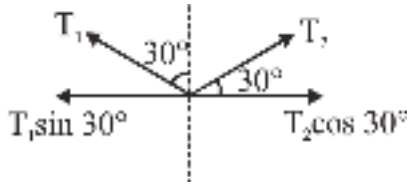
29) $a = \left(\frac{4-2}{4+2}\right)g$

$a = \frac{2}{6} \times 10 = \frac{10}{3}$

$30 - kx = 3 \times \frac{10}{3}$

kx = 20

$x = \frac{20}{100} = 0.2$




30)

$T_1 \sin 30° = T_2 \cos 30°$

$$\frac{T_1}{T_2} \Rightarrow \frac{\cos 30°}{\sin 30°} \Rightarrow \frac{\left(\frac{\sqrt{3}}{2}\right)}{\left(\frac{1}{2}\right)} \Rightarrow \frac{\sqrt{3}}{1}$$

31) $30 - T_2 = 3a$ .....(i)

$T_2 - T_1 = 6a$ .....(ii)

$T_1 - 10 = 1a$ .....(iii)

$20 = 10a$

$a = 2\ m/s^2$

32)

$10 - T_1 = 1 \times a$

$30 + T_1 - T_2 = 3a$

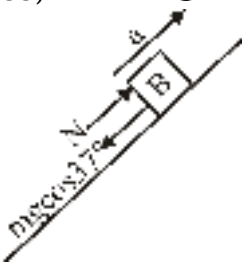
$T_2 - 30 = 3a$

We get

$a = \frac{g}{7} \downarrow$

and $T_1 = \frac{6g}{7}$

33) $a = \frac{21-12-6}{3} = 1\ m/s^2$



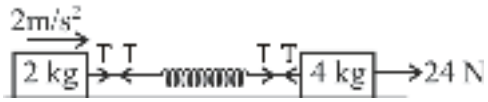

For block B :

$N - 6 = 1\ a$

$N - 6 = 1(1)$

$N = 7$ Newton

34)

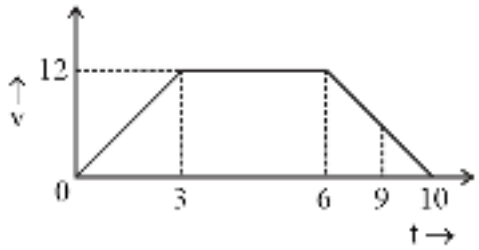



$T = 2(2)$ $\quad\quad$ $24 - T = 4\ (a)$

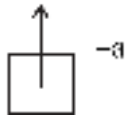
$T = 4N$ $\quad\quad$ $24 - 4 = 4a$

$a = 5\ m/s^2$

35) The fourth coin is under the weight of six coins above it. Hence,
Reaction of the $4^{th}$ coin on $5^{th}$ coin = Force on the $4^{th}$ coin due to $5^{th}$ coin = 6 mg




36)
at $8^{th}$ second, the object is retarding upwards.

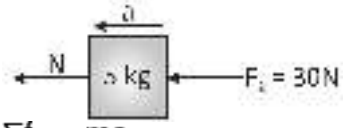



So,
N = mg – ma
$N = 1600(10) - 1600(3) \Rightarrow 1600(7)$
N = 11200 N

37)

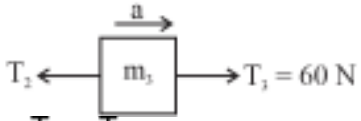
ΔP = Area under F - t Plot.
$= \frac{1}{2}(40 + 60)10$
= 500 N-s.

38) Acceleration of system, $a = \frac{\text{Net force}}{\text{Total mass}}$
$a = \frac{50-30}{10} = \frac{20}{10} = 2\ m/s^2$
$\boxed{a = 2\ m/s^2}$
Now, FBD of block - C



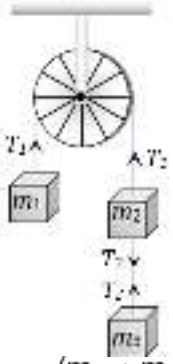

$\Sigma f_x = ma$
$\Rightarrow N - 30 = 5 \times a$
$\Rightarrow N = 5 \times 2 + 30 = 40\ N$

39) $a = \frac{F}{m_1 + m_2 + m_3} = \frac{60}{3+4+5} = 5\ m/s^2$
**FBD of block $m_3$:**




$\therefore T_3 - T_2 = m_3 a$
$\Rightarrow T_2 = 60 - 5 \times 5 = 60 - 25 = 35\ N$

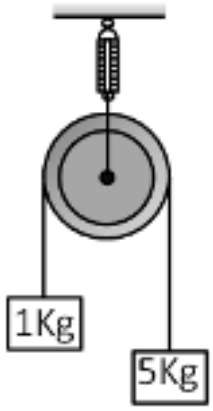
40) Tension in the string between $m_2$ and $m_3$ is given by



$$a = \frac{(m_2 + m_3 - m_1)}{(m_1 + m_2 + m_3)}g = \frac{g}{3}$$

$T_2 = m_3g - \frac{m_3g}{3} = \frac{4}{3}g = 13.06$ N

41) $T_1 : T_2 : T_3$
2g : 5g : 10g
2 : 5 : 10



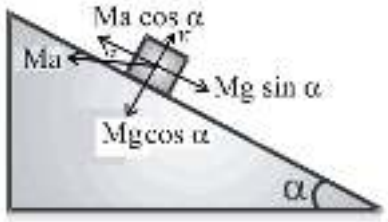

42)

$F = 2T;\ T = \frac{2 \times 5 \times 1}{5+1}g = \left(\frac{10}{6}g\right)$ N

$F = \left(\frac{10}{3}g\right)$ N

Reading = $\frac{10}{3}$ kg-wt

43) **Question Explanation :** As shown in given figure to find acceleration of incline surface so that block remain stationary.
**Concept :** Pseudo force.
**Solution :**




$F_{Net} = 0$
Ma cos α = Mg sin α
a = g tan α

**Final Answer :** Option (2)

44) Let n be the maximum number of bullets per second the person can fire. Since, force = rate of change of momentum

$\Rightarrow 144 = n \times \frac{40}{1000} \times 1200$

$\Rightarrow$ n = 3.

45)

**Explain Question :** Asking about the acceleration of the rocket 50 second after the blast.
**Concept :** This question is based on Rocket propulsion.
**Solution :**

$V_r \frac{dm}{dt} = M'a$ ...(1)

Mass after 50 sec.

$M' = M - \frac{dm}{dt} \times 50$

$M' = 1500 - 10 \times 50$

$= 1000$ kg

From eq. (1)

$5 \times 10^3 \times 10 = 1000 \times a$

$a = 50$ m/s$^2$

**Final answer :** Option (3)

CHEMISTRY

46) **Question Explanation:** The question asks to identify which of the following elements- Sodium (Na), Potassium (K), Nitrogen (N), or Oxygen (O)-possesses the highest lonization Potential (I.P.).

**Concept:** lonization Potential, Periodic Trends, Nuclear Charge, Half-filled Stability, Atomic Size.

**Solution:**

A. **Atomic Size:** I.P. increases across a period (left to right) and decreases down a group. Na and K (Group 1) have large radii and low I.P. Nitrogen and Oxygen (Period 2) have smaller radii and higher I.P.
B. **Electronic Configuration:** * N: $1s^2 2s^2 2p^3$ (Exactly half-filled p-orbital, highly stable)
    A. O: $1s^2 2s^2 2p^4$
C. **Comparison:** Removing an electron from Nitrogen's stable $2p^3$ configuration requires more energy than removing one from Oxygen's 2p configuration.

**Final Answer:** Option (3) (N)

47)

**Explain Question :** Arrange the lead atom and its positive ions by decreasing size.

**Concept :** This question is based on Ionic Radius.

**Solution :** Cations are always smaller than their neutral parent atoms. As positive charge increases, the nucleus pulls the remaining electrons closer, thus further decreasing the ionic size. So, the correct order is $Pb > Pb^{2+} > Pb^{4+}$

$$\text{Atomic/Ionic Radius} \propto \frac{1}{Z_{eff}} \propto \frac{\text{Negative charge}}{\text{Positive charge}}$$

**Final Answer :** option (2)

48)

Due to stable configuration N and inert gases these show endothermic addition of $e^-$.

49) **Question Explanation:** Identify bond with minimum polarity.

**Concept:** Electronegativity.

**Solution:** Bond polarity depends on ΔEN.
P (≈2.1) and H (≈2.1) have nearly equal electronegativity, so ΔEN ≈ 0, making P–H least polar among given bonds.

**Final Answer :** Option (3)

50) $$\text{E.N.} \propto \frac{\text{+ve charge}}{\text{–ve charge}}$$

51) **Question Explanation :**
Identify geometry not obtained from $sp^3d$ hybridization.

**Concept :**
This question is based on VSEPR

**Solution :**
$sp^3d$ hybridization given trigonal bipyramidal electron geometry, leading to see-saw, T-shape and linear shapes. Trigonal planar ($NO_3^-$) arises from $sp^2$, not $sp^3d$.

**Final Answer :**
Option (1)

52) **Explanation Question:**
Question is asking about sharing of electrons In Triple bond,
**Concept:**
This question is based on Covalent bond, valency
**Solution:**
In Triple bond, there is sharing of 6 electrons

**Final Answer: (3)**

53)

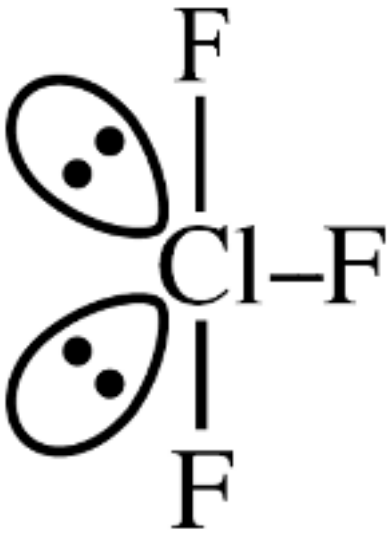
**Explain Question:** Why can Phosphorus form five bonds with chlorine but Nitrogen cannot?

**Concept:** Octet Expansion, Covalency.

**Solution:** Nitrogen is in Period 2 and has only s and p orbitals. It cannot exceed an octet. Phosphorus (Period 3) has vacant 3d-orbitals, allowing it to expand its octet.

**Final Answer:** Option (1)

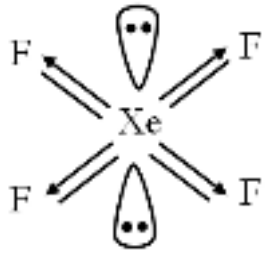
54) **Question Explanation :** Question is asking about the molecule that is T-shaped.
**Concept :** This question is based on VSEPR theory.
**Solution :**
The molecule that is T-shaped in $ClF_3$.




**Final Answer :** Option (3)

55)

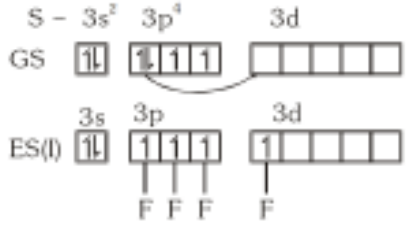
**Explain Question :-** Molecule that has square planar shape and is non-polar.
**Concept :-**This question is based on VSEPR
**Solution :-**
$XeF_4$: $sp^3d^2$




square planar
$\mu = 0$ dipole moment is zero.
**Final answer :-** (1)

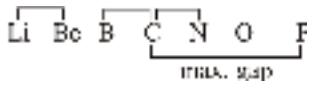
56) $NO_2^+$ = sp Hybridisation - linear
$CO_2$ = sp Hybridisation
$XeF_2$ = $sp^3d$ (2σ + 3 LP) - Linear

57)




4 Vacant Orbitals

58)




59) **Question Explanation :** Identify the element's electron configuration from its ionization energies.

**Concept:** This question is based on Ionization Energy.

**Solution:** The large energy jump between the second and third ionization energies indicates the element has two valence electrons. The configuration ending in $3s^2$ is the only option with two valence electrons.

**Final Answer:** Option (4)

60)

Concept

61) EN of S, C, I = 2.5
EN of H = 2.1

62) **Asking For:**
• Identify which of the given orbital overlaps are incorrect, assuming the z-axis is the internuclear axis.
**Concept:**

A. **Orbital Overlap:** The merging of atomic orbitals to form molecular orbitals, leading to bond formation
B. **Sigma (0) Bonds:** Formed by head-to-head overlap along the internuclear axis
C. **Pi (π) Bonds:** Formed by sideways overlap of p-orbitals above and below the internuclear axis
D. **Internuclear Axis:** The imaginary line connecting the nucler of two bonded atoms.
E. When z-axis is inter nuclear axis, Pz orbitals overlap head to head, and Px and Py overlap sideways

**Solution/Explanation:**

A. **(a) $2P_y + 2P_y \rightarrow \pi$ bond:**
   A. Since the z-axis is the internuclear axis, $P_y$ orbitals overlap sideways, forming a π bond. This is correct.
B. **(b) $2P_x + 2P_x \rightarrow \sigma$ bond:**
   A. Since the z-axis is the intemuclear axis, $P_x$ orbitals overlap sideways, forming a π bond, not a σ bond. This is incorrect
C. **(c) $2P_z + 2P_z \rightarrow \pi$ bond:**
   A. Since the z-axis is the internuclear axis, $P_z$ orbitals overlap head-to-head, forming a σ bond, not a π bond. This is incorrect.
D. **(d) $s + P_z \rightarrow \pi$ bond:**
   A. s orbitals and $P_z$ orbitals overlap head-to-head along the internuclear axis, forming a σ bond, not a π bond. This is incorrect
E. **(e) $s + P_z \rightarrow \sigma$ bond:**
   A. s orbitals and $P_z$ orbitals overlap head-to-head along the internuclear axis, forming a sigma bond. This is correct.

Therefore, the incorrect overlaps are (b), (c), and (d).
**Answer:**
**Option: 2**

63) Theory based

64) $[N_2] = \frac{1.4}{28 \times 2}$, $[H_2] = \frac{1}{2 \times 2}$

$[N_2] : [H_2] = \frac{1}{40} : \frac{1}{4}$

= 1: 10

65) $K_P = P_{CO_2}$

66)

**Solution:**

A. **Option 1: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$**
   A. $\Delta n_g = 2 - (1 + 3) = -2$
   B. $Kp \neq Kc$
B. **Option 2: $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$**
   A. $\Delta n_g = 2 - (1 + 1) = o$
   B. Kp = Kc
C. **Option 3: $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$**
   A. $\Delta n_g = (2 + 1) - 2 = 1$
   B. $Kp \neq Kc$
D. **Option 4: $N_2O_4(g) \rightleftharpoons 2NO_2(g)$**
   A. $\Delta n_g = 2 - 1 = 1$
   B. $Kp \neq Kc$

**Final Answer :** options (2)

67) **Solution :**

$Kp = K_c (RT)^{\Delta n_g}$

$$\log \frac{K_p}{K_c} = \Delta n_g \log RT$$

Given : $\log \frac{K_p}{K_c} + \log RT = 0 \Rightarrow \log \frac{K_p}{K_c} = -\log RT$

$\Delta n_g = -1$

Reaction : $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

$\Delta n_g = 2 - (2 + 1) = -1$

**Final Answer:** (2)

68) **Concept :** Keq ∝ stability of product

**Solution :** In option (4), value of equilibrium constant(k) is minimum, so products are less stable.

**Final Answer :**

$XeO + \frac{1}{2}O_2 + F_2 \rightarrow XeO_2F_2$, $K_4 = 1.4 \times 10^{-3}$

69) **Solution:**

$$K_{eq} \propto \frac{[Product]}{[Reactant]}$$

$K_{eq} \rightarrow$ Very low, more reactant

$K_{eq} \rightarrow$ Very high, more product

$K_{eq} = 10^{-14} \Rightarrow$ Very low, mostly reactant present in equilibrium mixture.

**Final Answer:** Option(1)

70) **Solution:**

$$\underset{(g)}{Br_2} \rightleftharpoons \underset{(g)}{2Br}$$

$T_1 = 500$ K, $K_1 = 10^{-10}$

$T_2 = 700$ K, $K_2 = 10^{-5}$

$\square$ T ↑ ⇒ K ↑ (Endothermic Reaction)

for endothermic Reaction ⇒ K ∝ T

**Final Answer: (1)**

71) **Question Asking:** It ask about the value of $K_P$.

Given: $NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$ ($K_C = 4$) ....(1)

T = 527 + 273 = 800 K

**Concept:**

$K_P = K_C(RT)^{\Delta ng}$

To get the required reaction, we have to reverse & multiply the equation by × 2

$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + H_2(g)$ $K_C = 4$

an reversing & multiplying by 2

$N_2(g) + 3H_2(g) \rightleftharpoons 2\ NH_3(g)$

$K_C' = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$

**Calculation:**

$K_P = K_C (RT)^{\Delta ng}$

$= \frac{1}{16} \times [R \times 800]^{-2}$

$K_P = \left(\frac{1}{4 \times 800R}\right)^2$

**Option (3) is correct answer**

72)

**Question Explanation:** Check correctness of Statement-I and Statement-II for equilibrium: $H_2O(s) \rightleftharpoons H_2O(\square)$

**Concept:** Le-Chatelier's principle

**Solution:**

• Melting of ice is **endothermic**.

• Ice has **greater volume** than liquid water.

• Increase in temperature favors **endothermic** direction.

• Increase in pressure favors the side with **smaller volume**.

Statement-I: Increasing temperature favors melting and increasing pressure favors liquid water (smaller volume) → **more water forms** → **correct**.

Statement-II: Forward reaction is endothermic (**true**), but it says volume of water is greater than ice (**false**, ice has greater volume). Hence overall **incorrect**.

Statement I is correct but Statement II is incorrect.

**Final Answer:** (Option 2)

73) **Mathematical Calculation:**

**1. Calculate moles of HI dissociated:**

A. Moles of HI dissociated = Initial moles of HI x Degree of dissociation
B. Moles of HI dissociated = 8 moles x 0.25 = 2 moles

**2. Calculate moles of $H_2$ and $I_2$ formed:**

A. From the balanced equation, 2 moles of HI dissociate to produce 1 mole of $H_2$ and 1 mole of $I_2$.
B. Since 2 moles of HI dissociated, 1 mole of $H_2$ and 1 mole of $I_2$ are formed.

**E. Final Answer:**

A. Moles of $I_2$ at equilibrium = 1 mole

74)

$[H^+] = 6 \times 10^{-4}$ M

$pH = -\log[H^+]$

$= -\log(6 \times 10^{-4})$

$= 4 - \log 6$

$= 4 - 0.78$

= 3.22

75)

Ostwald's dilution law is valid only for weak electrolytes because strong electrolytes are almost completely ionized and do not obey this law.
**Solution:**
• $HNO_3$ → strong acid (strong electrolyte)
• $H_2SO_4$ → strong acid (strong electrolyte)
• NaOH → strong base (strong electrolyte)
• HCN → weak acid (weak electrolyte) → follows Ostwald dilution law
**Final Answer:** Option (4)

76) $K_a = \frac{C\alpha^2}{1-\alpha}$
$\because 1 - \alpha \approx 1$
$\Rightarrow K_a = C\alpha^2$
$10^{-5} = 10^{-1}\alpha^2$
$\alpha^2 = 10^{-4}$
$\alpha = 10^{-2}$

77)

Due to common ion effect addition of $NH_4Cl$ to $NH_4OH$ decreases the ionization of $NH_4OH$.

78)

Basic strength ↑, $K_b$ ↑, $pK_b$ ↓
Basic strength ↓, $K_b$ ↓, $pK_b$ ↑

79) **Solution :**
At this temperature, $pK_w = 13.4$
Neutral pH = 13.4/ 2 = 6.7
Measured pH = 7, which is greater than 6.7
Therefore, the solution is slightly basic.
**Final answer :** 2

80)

$H_2O \rightleftharpoons H^+ + OH^-$ $\Delta H$ = +ve
T↑, forward, $[H^+]$↑, $[OH^-]$↑
pH↓, pOH↓

81) **Solution :**
Increasing temperature increases ionization of water, so $K_w$ increases.
Adding $H^+$ or $OH^-$ changes pH but does not change $K_w$.

Pressure has negligible effect.
**Final Answer :** (4)

82) **Solution :**
**Statement I :** True → $CH_3COOH$ ionises more in $H_2O$ because ions are better stabilised
**Statement II :** True → $H_2O$ has higher dielectric constant than $D_2O$, so more polar.
Statement II correctly explains statement I
**Final Answer :** Option (1)

83) **Solution :**
**1. $HClO_4 + KClO_4$ :**
$HClO_4$ is strong acid → fully ionized.
$ClO_4^-$ from $KClO_4$ does not suppress $HClO_4$ (already strong)
**2. $H_2SO_4 + KOH$ :**
$H_2SO_4$ strong acid, KOH strong base → neutralization, not common ion effect
**3. $CH_3COOH + HCl$ :**
$CH_3COOH$ + HCl : is weak acid, HCl is strong acid providing $H^+$ → suppresses $CH_3COOH$ ionization
Only option 3 demonstrate common ion effect.
**Final answer :** 3

84)

$[H^+] = 10^{-6}$ so pH = 6
and $[OH^-] = 10^{-6}$ so pOH = 6
pH + pOH = 12
So $pK_w = 12$
$K_w = 10^{-12}$

85)

On heating water $K_w$ increases and thus pH scale for neutrality changes from 7 to some lower value, depending upon $K_w$ value.

86) **Solution :**
$[NaOH] = [OH^-] = \frac{0.2 \times 1000}{40 \times 100} =$
$[OH^-] = 5 \times 10^{-2}$
$pOH = -\log[OH^-] = -\log(5 \times 10^{-2})$
$pOH = 2 - \log 5 = 2 - 0.69 = 1.31$
$pH = 14 - pOH = 14 - 1.31 = 12.69$
**Final Answer:** option (3)

87)

**Explain the Question :** We need to find pH of a very dilute acidic solution like $10^{-7}$ M HCl.

**Concept :** This question is based on pH of very dilute solution.
**Solution :**
For very dilute solution :
$[H^+]_{total} = [H^+]_{acid} + [H^+]_{H_2O}$
$[H^+]_{total} = 10^{-7} + 10^{-7} = 2 \times 10^{-7}$
pH = – log $[H^+]$ = –log $(2 \times 10^{-7})$
pH = 7 – 0.3 = 6.7 (between 6 and 7)
**Final Answer:** Option (4)

88) $N_1V_1 + N_2V_2 = NV$
$4 + 1 = W \times 300$
$[H^+] = \frac{5}{300} = 1.6 \times 10^{-2}$
$P^H$ = 2–log1.6 = 1.78

89) **Solution :**
$pH = -\log\left[\frac{50 \times 0.4 - 50 \times 0.2}{100}\right] = 1$
**Final Answer :** (3)

90) **Explaining :** Choose the correct statements
**Concept :** This question is based on multiple concepts.
**Solution :**
(a) Equivalence of strong acid = 0.1 × 2 × 400 = 80
Equivalence of strong base = 0.1 × 400 = 40
 $[H]^+$ of mixture = $\frac{80-40}{800} = \frac{1}{20}$
 pH = –log$[H^+]$ = –log $\left(\frac{1}{20}\right)$ = 1.3
**Final Answer : (2)**

BIOLOGY

91) **Question explanation**:
To identify the incorrect statement regarding uses and products of algae.
**Concept**:
Economic importance of algae
**Solution**:
Algae release oxygen during photosynthesis → increase dissolved $O_2$ (correct).
Algin is obtained from brown algae and carrageenan from red algae, but the statement reverses them → incorrect.
Agar is obtained from *Gelidium* and *Gracilaria* (correct).
*Laminaria* and *Sargassum* are edible brown algae (correct).
**Final Answer**:
Option (2)

92) NCERT Pg. # 27

93)
**Correct Answer: 4. Cycas**
**Explanation :** Cycas (a gymnosperm),Gametophyte is dependent on the sporophyte,Male gametophyte develops within the pollen grain and Female gametophyte develops within the ovule.

94) NCERT XI pg.#35

95) NCERT XI pg.#35

96)

**Explain Question :** The question asks to identify two plants, labeled (a) and (b), from a provided image. Both plants are examples of pteridophytes.
**Concept :** This question is based on Pteridophytes
**Explanation :** Figure (a) shows a plant with a creeping rhizome and upright, branching stems covered in small, scale-like leaves. This morphology is characteristic of Selaginella.
Figure (b) shows a plant with a distinctly segmented, jointed stem and whorls of leaves at each node. This structure is unique to Equisetum.
**Final answer :** 3

97)

**Explain Question :** The question wants to know the type of pteridophytes Selaginella and Salvinia are, based on their spore production.
**Concept :** This question is based on Pteridophyta, General Characters, Life Cycle And Heterospory
**Explanation :** "Selaginella and Salvinia are heterosporous pteridophytes, producing both microspores and megaspores."
**Final answer :** 2

98) NCERT XI Pg # 38, 39

99)

XI New NCERT Page No. 33

100) NCERT ($XI^{th}$) Pg. # 30

101) **Question Explanation :** This question is asking about difference between Bryophytes and Pteridophytes
**Concept :** This question  is based on difference between Bryophytes and Pteridophytes
**Solution :** Bryophytes have a dominant gametophyte generation, and their sporophyte is

dependent on the gametophyte for nutrition and support. In contrast, pteridophytes have a dominant and independent sporophyte generation, which is the more conspicuous plant body, while the gametophyte is typically small and short-lived but also independent.
**Final Answer :** Option (2)

102)

NCERT XI, Pg # 29

103) NCERT (XI) Pg. # 32

104) **Explain Question :** The question asks to identify the correct terms to fill in these blanks based on the characteristics of each algal class.
**Concept :** This question is based on classes of algae
**Explanation :**
**A (Major Pigments of Rhodophyceae):** The major pigments in Rhodophyceae are Chlorophyll a and d.
**B (Stored Food of Phaeophyceae):** The stored food in Phaeophyceae is Mannitol and Laminarin.
**C (Cell Wall of Chlorophyceae):** The cell wall of Chlorophyceae is made of Cellulose.
**D (Flagellar Number of Rhodophyceae):** Flagella are absent in Rhodophyceae.
**Final answer :** 2

105) NCERT-XI Pg # 36

106)

**Explain Question :** The question requires identifying the two correct statements from a given set of four about the characteristics of gymnosperms

**Concept :** This question is based on Gymnosperm, General Characters And Life Cycle

**Solution :**

**(A) Diplontic life cycle is found in all seed bearing plants:**
This statement is correct. All seed-bearing plants (gymnosperms and angiosperms) exhibit a diplontic life cycle where the dominant phase is the diploid sporophyte.

**(B) In Pinus, both male cones and female cones are borne on the same tree:**
This statement is correct. Pinus is monoecious, meaning both male (pollen) cones and female (seed) cones are present on the same individual plant.

**Final Answer :** option (1)

107) NCERT (XI) (Eng.) Pg. # 219 ($2^{nd}$ para)

108) NCERT XI Pg.#220

109) NCERT-XI, Pg#219

110) NCERT XI, Pg # 215 (4th para)

111) NCERT ($XI^{th}$) (E) Pg. # 215 ($1^{st}$ Para)
NCERT ($XI^{th}$) हिन्दी पेज # 215 (दूसरा पेराग्राफ)
[हिन्दी NCERT में सिर्फ विसरण लिखा है, सुसाध्य विसरण नहीं इसलिए Eng. NCERT से confirm करें।]

112) NCERT Page No. # 220

113)

NCERT-XI, Page No.-212

114)

Correct answer: 4. a-(ii), b-(iii), c-(i)

Explanation:

A. (a) Jan Ingenhousz → (ii) Essentiality of sunlight to plants
   He showed that plants need sunlight to purify air (photosynthesize).
B. (b) Joseph Priestley → (iii) Discovered oxygen
   He discovered that plants release a gas (later identified as oxygen) which supports combustion.
C. (c) T. W. Engelmann → (i) First action spectrum of photosynthesis
   He used a prism and green algae with bacteria to show that blue and red light are most effective in photosynthesis.

So the correct matching is:
a-(ii), b-(iii), c-(i)

115) **Question explanation**:
To identify what is demonstrated by Moll's half-leaf experiment
**Concept**:
Photosynthesis.
**Solution**:
In Moll's half-leaf experiment, part of the leaf is deprived of $CO_2$ while the rest is exposed. Only the exposed part forms starch, proving that $CO_2$ is necessary. It does not test chlorophyll directly.
**Final Answer**:
Option (2)

116) **Explain Question :** The question asks you to match four photosynthetic pigments with

their corresponding colors as they would appear after being separated on a chromatogram.
**Concept :** This question is based on How Many Pigments Are Involved In Photosynthesis
**Explanation :**
**A. chl-a**
corresponds to i. Bright or blue-green
**B. chl-b**
corresponds to iii. Yellow-green
**C. xanthophyll**
corresponds to ii. Yellow
**D. carotenoids**
corresponds to iv. yellow to yellow orange
**Final answer :** 2

117) **Question Explanation:** Identifying the wrong match in photosynthesis.
**Concept:** Photosynthesis processes
**Solution:** The **PS-II** (Photosystem II) reaction center has a maximum absorption peak at 680 nm, so it is designated as **P680**. **P700** is the reaction center for **PS-I** (Photosystem I). Therefore, the match between PS-II and P700 is incorrect.
**Final Answer:** PS-II — P700

118)

**Question Explanation:** Asking the saturation $CO_2$ concentration of $C_4$ and $C_3$ plants.

**Concept:** Factors affecting photosynthesis - Carbon dioxide.

**Solution:**

The saturation of $CO_2$ concentration for $C_4$-plants are $360\ \mu L\ L^{-1}$ and the saturation point of $C_3$-plants is $450\ \mu L\ L^{-1}$.

**Final answer :** Option (2)

119)

**Explain Question –** The question asks which statement is not related to C4 plants.

**Concept –** Based on C4 photosynthesis.

**Solution –** C4 plants show Kranz anatomy, double $CO_2$ fixation, and adaptation to dry tropical conditions. Sugar synthesis occurs in bundle sheath cells, not in mesophyll cells, so option 3 is incorrect.

**Final answer – Option 3**

120) NCERT XI Page # 217

121) NCERT Pg.# 214

122) **Question Explanation:** Identify the correct matches between $C_3$ and $C_4$ plants.
**Concept:** $C_3$ vs $C_4$ pathway.
**Solution:**
I. Primary $CO_2$ acceptor → $C_3$: RuBP, $C_4$: PEP → Given RuBP for both → Incorrect
II. Primary product in $C_3$ cycle → PGA in both cases → Correct
III. Carbon number of acceptor → $C_3$: 5C (RuBP), $C_4$: 3C (PEP) → Given reversed → Incorrect
IV. Initial carboxylation → $C_3$: mesophyll, $C_4$: mesophyll (not bundle sheath) → Incorrect
Correct = II only
**Final Answer:** Option (2)

123)

NCERT Pg # 217 (Fig. 13.8)

124)

Question Explanation: The question asks to identify the correct statements about the structure, position, and dimensions of human kidneys.

Concept: The question is based on anatomical features of kidneys.

Solution: The correct answer is option 3.

A. Statement (A) is correct: The medulla is divided into conical masses called renal pyramids that project into the calyces.

B. Statement (B) is correct: Cortex extends between the medullary pyramids as renal columns called columns of Bertini.

C. Statement (C) is incorrect: Kidneys are located close to the dorsal (back) wall of the abdominal cavity, not ventral (front).

D. Statement (D) is correct: Each kidney measures 10–12 cm in length, 5–7 cm in width, 2–3 cm in thickness, with average weight 120–170 g.

Final answer: Option 3

125) **Question Explanation:** Primary function of sweat glands.
**Concept:** Thermoregulation
**Solution:** The primary function of sweat glands is to regulate body temperature. When the body gets too hot, sweat is produced and released onto the skin. The evaporation of this sweat absorbs heat from the body's surface, creating a **cooling effect**. While sweating also removes some wastes and contributes to osmoregulation, these are secondary to its main role in thermoregulation.
**Final Answer:** To facilitate a cooling effect on the body surface.

126)

NCERT-XI, Pg. # 213

127) NCERT-XI, Pg. # 214

128) **Question Explanation**: Identify the condition indicated by the presence of both glucose and ketone bodies in urine.

**Concept**: The question is based on excretory disorders.

**Solution**:

A. The presence of **glucose (glycosuria)** and **ketone bodies (ketonuria)** in urine occurs due to impaired carbohydrate metabolism, typically when insulin is deficient.
This condition indicates the sign of Diabetes mellitus.

**Final answer**: Option 3

129) Aldosterone: This hormone, secreted by the adrenal cortex, acts on the kidneys to increase the reabsorption of sodium ions (Na+) and water. This leads to an increase in blood volume and blood pressure.

130)

**Question Explanation**: This question is about the **reabsorption mechanisms** of different substances in the nephron (kidney tubules).

**Concept**: **Urine formation**

**Solution**: The correct answer is **option 1: A = Actively; B = Passively**

A. Glucose and amino acids are **essential nutrients**, so they are **actively reabsorbed** using energy.
B. Nitrogenous waste products like **urea** are **passively reabsorbed** along the concentration gradient.

**Final answer**: **Option 1**

131) NCERT-XI, Pg. # 206

132)

**Explain Question :** Correct statements is /are ?

**Concept :** This question is based on Mechanism of urine formation.

**Solution :**
(a) Vasa recta is absent or highly reduced in cortical nephrons — Correct
(b) Ammonia is less toxic than urea — Incorrect
(c) GFR is equal to 125 me/minute — Correct
(d) PCT also helps to maintain the pH and ionic balance of the body fluid — Correct

**Final Answer :** option (3)

133)

**Explain Question :** Glucose and 75% water is absorbed in ?

**Concept :** This question is based on Selective reabsorption.

**Solution :** Glucose and 75% water is absorbed in Proximal convoluted tubule.

**Final Answer :** option (1)

134)

**Question Explanation :** The role of kidneys in osmoregulation.

**Concept :** Osmoregulation.

**Solution :** The kidneys are the primary organs for maintaining blood osmotic concentration (osmoregulation). They do this by adjusting water reabsorption to produce either dilute (hypotonic) or concentrated (hypertonic) urine as needed by the body.

**Final Answer :** (1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

135)

**Solution/Explanation/Calculation:**

Here's why:

A. Ammonotelic: Animals that excrete nitrogenous waste primarily in the form of ammonia. Examples include aquatic animals like fish.

B. Ureotelic: Animals that excrete nitrogenous waste primarily in the form of urea. Examples include mammals and some amphibians.

Uricotelic: Animals that excrete nitrogenous waste primarily in the form of uric acid. Examples include birds and reptiles.
**The correct answer is 3. The types of nitrogenous waste produced by various classes of vertebrates.**

136) **Question Explanation:**
To identify the false statement
**Solution Explanation:**
I. Outer cortex and inner medula are the two zones in kidney. This statement is true.
II. Medulla is divided into about 8 to 12 medullary pyramids. This statement is true.
III. Pyrmid projects into calyx. This statement is true.
IV. Inwards extension of cortex between the pyramids is called renal column of Bertini. This stetement is true
**Final Answer:**
Since all four statements (I, II, III, and IV) are true

137)

**Explanation :**

I. The composition of the filtrate would be most like plasma in the tubule.

A. Filtration occurs in Bowman's capsule (A), where the filtrate closely resembles plasma without proteins.
I → A

II. The urine would be most concentrated in the collecting duct.

A. The collecting duct (E) is where final water reabsorption occurs under ADH influence, making the urine most concentrated.
II → E

III. Most of the glomerular filtrate is reabsorbed into peritubular capillary.

A. This occurs primarily in the proximal convoluted tubule (B), where ~65–70% of filtrate is reabsorbed.
III → B

**Correct Answer: Option 4 → I-A, II-E, III-B, IV-E, V-B**

A. IV. Conducting of urine to the pelvis of the kidney from the structure.

A. The collecting duct (E) transports urine to the renal pelvis.
IV → E

B. V. Most water is reabsorbed by the structure.

A. The loop of Henle (D), especially the descending limb, is responsible for most water reabsorption.
V → D

138)

**Question explanation:** Identify the correct sequence(s) related to hormonal and renal regulation.
**Concept:** Excretory system – Hormonal regulation

**Solution:**

A. **An increase in body fluid volume → switch off the osmoreceptors → suppresses the ADH release** - Correct – When fluid volume is high, osmoreceptors are inhibited, reducing ADH secretion, leading to less water reabsorption.

B. **ADH-Constricting effect on blood vessel → B.P. high → Glomerular blood flow more → GFR more** - Correct – ADH has vasoconstrictor action (called vasopressin effect), raising blood pressure, which increases glomerular blood flow and GFR.

A. **Angiotensinogen → Angiotensin I → Angiotensin-II → Adrenal cortex → Aldosterone** - Correct – This is the **RAAS pathway** (Renin-Angiotensin-Aldosterone System), regulating sodium and water retention.

Since all three sequences are correct → **Option 4 (All)**

**Final answer:** Option 4

139) **Question Explanation :**
What is the correct sequence of steps in the dialysis process?

**Concept :**
This question is based on Dialysis.

**Solution :**
Let’s analyze the correct order:
D. Blood is drained from a convenient artery → Start of process
B. Blood is mixed with heparin to prevent clotting
E. Blood is passed through cellophane tube in dialysis fluid
F. Nitrogenous wastes are removed by diffusion
C. Blood is mixed with anti-heparin before return

A. Blood is returned into a vein

**Hence,**
**Option (3) is correct :** D → B → E → F → C → A

140) **Question Explanation**: Determine which compound increases in excretion after high protein intake.

**Concept**: The question is based on nitrogen metabolism and excretory products in humans.

**Solution**:

A. Proteins are broken down into amino acids, which undergo deamination.This produces ammonia, which is toxic and is converted into **urea** in the liver.
B. Urea is then excreted by the kidneys.So, high protein intake leads to increased **urea** production and excretion.

**Final answer**: Option 1

141)

Class XI Ncert page 209.

142) **Question Explanation:** Find correct about Angiotensin II
**Concept :** RAAS
**Solution :**
Angiotensin II → act as Vasoconstrictor. (v)
→ Increases B.P. (ii)
→ Increases GFR (i)
→ it also Increases aldosterone secretion (iv)
Which help in Na retention.
**Final Answer :** option (3). Four

143)

NCERT Pg. No. 209

144) **Question Explanation :**
The question is asking for the correct formula to define Net Filtration Pressure (NFP) in the kidneys during glomerular filtration.

**Concept :**
Mechanism of urine formation.

**Solution :**
NFP = GHP – (BCOP + CHP)
This formula represents the net force driving fluid out of the glomerulus into Bowman's capsule.

**Final Answer :**
Option (3)

145) **Question Explanation :** This question asks about reabsorption and secretion of major substances at different parts of the nephron are shown in the given figure. In this figure A, B, C and D are respectively.

**Concept :**This question is based on Mechanism of urine formation.

**Solution :**
A: NaCl(Significant reabsorption in the ascending limb and PCT)
B: urea (some reabsorption and secretion throughout, especially in the collecting duct and PCT)
C: $K^+$ (Secretion in the DCT)
D: Ammonia (secretion in the DCT and PCT)

**Final Answer :** Option (2)

146)

**Question Explanation:**
Asks for the initial step in the process of urne formation within the kidneys.

**Solution/Explanation:**
Ultrafiltration (Glomerular Filtration): This is the first step occurring in the glomerulus. Blood pressure forces water and small solutes (like glucose, amino acids, ions, and urea) from the blood into Bowman's capsule, forming the glomerular filtrate.
Therefore, the first step is ultrafiltration.

**The correct answer is: option (1) Ultrafiltration.**

147)

NCERT-XI, Pg. # 212

148) **Question Explanation :**
The question is asking about the approximate osmolarity found in the inner medulla of the kidney in humans.

**Concept :**
This question is based on Counter-current mechanism.

**Solution:**
In the renal medulla, osmolarity increases from about 300 mOsmol/L at the cortex to a maximum in the inner medulla due to the counter-current multiplier and urea recycling. In humans, this peak osmolarity is approximately 1200 mOsmol/L.

**Final Answer :**
Option (3)

149) Class XI Ncert page 208.

150) NCERT (XIth) Pg. # 300

151)

**Explain Question :** Which play a significant role in the maintenance of medullary osmolarity gradient?

**Concept :** This question is based on Counter current Mechanism.

**Solution :** Counter current mechanism plays an significant role in the maintenance of medullary interstitial osmolarity gradient.

**Final Answer :** option (3)

152) Question Explanation : The question asks to identify the correct labeling of nephron parts (renal corpuscle region) in the diagram.
Concept : The question is based on structure of nephron
Solution : The correct answer is Option 1.

A. A = Afferent arteriole → brings blood into the glomerulus.
B. B = Efferent arteriole → carries blood away from the glomerulus.
C. C = Bowman's capsule → cup-shaped structure surrounding the glomerulus.
D. D = Proximal convoluted tubule (PCT) → continuation of Bowman's capsule leading into nephron tubule.

So, the correct matching is: A = Afferent arteriole, B = Efferent arteriole, C = Bowman's capsule, D = PCT.

Final answer : Option 1

153)

NCERT–XI Pg No - 213

154)

**Explain Question :** Which structure have stretch receptors for micturition?

**Concept :** This question is based on Micturition

**Solution :** The wall of urinary Bladder Have stretch Receptors for micturition.

**Final Answer :** option (1)

155)

**Question Explanation:** Identify the substances produced by the kidney.

**Concept:** Endocrine functions of kidney

**Solution:**

A. Renin → Secreted by juxtaglomerular (JG) cells; regulates blood pressure (RAAS).

B. Erythropoietin → Stimulates RBC production in bone marrow.

C. Calciferol (Vitamin $D_3$**)** → Not produced directly by kidney; the kidney converts it into active calcitriol, but does not synthesize calciferol itself.

Thus, both renin and erythropoietin are produced by the kidney.

**Final Answer:** Option 4

156) **Explain Question**: This question is about identifying the **part of the kidney** that serves as the gateway for the **ureter**, **nerves**, and **blood vessels**.
**Concept**: **Kidney anatomy**
**Solution**: The correct answer is **option 1: Hilus/Hilum**.

A. The **hilum** is the indentation on the **kidney's medial side** where structures such as the **ureter**, **renal artery**, **renal vein**, and **nerves** enter and exit the kidney.

**Final Answer**: **1.**

157) **Question Explanation:** Identifying animal germ layers.
**Concept:** Germ Layers
**Solution:** Figure (a) shows an animal with two germ layers, an outer ectoderm and an inner endoderm, separated by a non-cellular layer (mesoglea). This is a characteristic of diploblastic animals. Figure (b) shows an animal with three distinct germ layers: the outer ectoderm, the inner endoderm, and a middle mesoderm. This is a characteristic of triploblastic animals.
**Final Answer:** (a)-Diploblastic, (b)-Triploblastic

158) NCERT, Pg. # 42, Para-4.2.3

159)

NCERT page no.49

160) **Explaining :**
**Concept :**
**Solution :** Aschelminthes (Nematoda, e.g., Ascaris) are characterized by the presence of a pseudocoelom (false coelom), where the body cavity is not lined by mesoderm. They are commonly known as roundworms.
**Final Answer : (2)**

161) **Explaining :**
**Concept :**
**Solution : I. Incorrect:** Platyhelminthes are triploblastic but are acoelomate (lack a coelom). Aschelminthes are triploblastic but pseudocoelomate.
**II. Correct:** Cnidarians (e.g., *Hydra*, jellyfish) and adult Echinoderms (e.g., starfish) both exhibit radial symmetry.
**III. Correct:** Pseudocoelomates (e.g., nematodes/roundworms) are bilaterally symmetrical.
**Final Answer : (2)**

162) NCERT Page No. 56

163) NCERT Page No. 44

164) Question is Asking About: $NH_3$ is
Concept: Excretory matter
Solution/Explanation: $NH_3$ is highly toxic among all excretory products.
Hence correct answer is 3.

165) **Question Explanation**: Identifying the incorrect statement about sponges
**Concept**: Phylum Porifera (Symmetry)
**Solution**: Statement 2 is incorrect. **Sponges** are primarily **asymmetrical**, although a few (like *Sycon*) exhibit radial symmetry. Radial symmetry is characteristic of Cnidarians, Ctenophores, and adult Echinoderms.
**Final Answer**: option (2)

166) **Question Explanation**: Water entry point in Porifera.
**Concept**: Porifera Canal System.
**Solution**: In sponges (Porifera), water enters the body through numerous small pores all over the body surface called **ostia** and exits through the larger osculum.
**Final Answer**: option (2)

167) NCERT Pg # 53, 54

168)

**Question Explanation:**

Identifying key factors for Arthropod terrestrial success.

**Concept:**

Arthropod Adaptations

**Solution:**

The success of Arthropods as terrestrial invaders is primarily attributed to two major features: the chitinous exoskeleton, which provides physical protection and, crucially, prevents water

loss (desiccation) in dry environments, and the presence of wings (in most insects), which allowed them to disperse rapidly and occupy diverse ecological niches. While jointed appendages are a characteristic feature, wings and the exoskeleton are considered the most critical for their widespread success on land.

**Final Answer:**

Chitinous exoskeleton, wings, compound eye is present

169) NCERT Pg. # 47, 50, 51

170) NCERT-XII, Pg. # 47, Para 4.1.2, Figure 4.1 (a)

171) **Question Explanation:** Life cycle of a cnidarian.
**Concept:** Metagenesis
**Solution:** The diagram illustrates **metagenesis**, the alternation of generations between the polyp and medusa body forms in some cnidarians like **Obelia**. The **polyp** is the sessile, asexual form that produces the free-swimming medusa **asexually (a)**. The **medusa** is the sexual form that reproduces sexually to produce the polyp **(b)**.
**Final Answer:** a-asexually, b-sexually

172) **Solution/Explanation/Calculation:**

| ANIMAL | *Ornithorhynchus* (Duck billed platypus) | *Macropus* (Kangaroo) | *Pteropus* (Flying fox) | *Balaenoptera* (Whale) |
|---|---|---|---|---|
| OVIPAROUS/ VIVIPAROUS | Oviparous (Egg laying mammal) | Viviparous | Viviparous | Viviparous |
| PINNA | Absent | Present | Present | Absent |
| CLOACA / ANUS | Cloaca present | Anus present | Anus present | Anus present |
| |  |  |  |  |

173) **Question Explanation**: Identify the feature found only in mammals

**Concept**: Mammalia unique characteristics

**Solution:** The muscular diaphragm is a unique characteristic of mammals, separating the thoracic and abdominal cavities and aiding in respiration.

**Final Answer:** option (4)

174) **Solution/Explanation:**
A tympanum, or eardrum, is a thin membrane that vibrates in response to sound waves. It is a crucial part of the hearing process in many animals
In the given options birds, frogs and lizards have tympanum.
Tympanum is absent in snakes.

Therefore, the correct answer is option (4)

175) **Question Explanation:** Relationship between radiata and radial symmetry.
**Concept:** Body Symmetry
**Solution:** Both the Assertion and Reason are true and the Reason is a correct explanation. The term 'radiata' refers to a group of animals with radial symmetry. This body plan, where any plane through the central axis divides the organism into identical halves, is a key characteristic of phyla like Coelenterates and Ctenophores.
**Final Answer:** Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

176) **Question Explanation:** Identifying a phylum from its body features.
**Concept:** Phylum Mollusca Characteristics
**Solution:** The characteristics described, including an unsegmented body with a calcareous shell, and a body divided into a head, muscular foot, and visceral hump, are all defining features of the phylum **Mollusca**. The excretory system featuring one or two pairs of kidneys, often referred to as Keber's organs, is also a key trait of this phylum.
**Final Answer:** Mollusca

177)

**Question Explanation**
The question is asking which animals are classified as jawless fishes (Cyclostomes).

**Concept**
Chordata – Cyclostomata

**Solution**
Jawless fishes lack jaws and paired fins. Hagfish and lamprey are the only examples of jawless fishes. Eel, rohu, magur, flying fish, and dog fish have jaws.

**Final answer: Option 4**

178) NCERT XI, Pg # 58

179) NCERT Pg. # 52, Para-4.2.6

180) NCERT Pg. # 52